

44. Octal Serial Peripheral Interface (OSPI)

This is the OSPI_B version of the OSPI peripheral module.

Flash memory ISSI IS25WX064-JWLE is packaged for SIP product.

OSPI_B is referred to as OSPI in this chapter.

44.1 Overview

The xSPI (eXpanded Serial Peripheral Interface) protocol specifies the interface for nonvolatile memory devices, which provides high data throughput, low signal count, and limited backward compatibility with legacy SPI devices. The electrical interface can deliver up to 333 MB per second raw data throughput. The OSPI is compliant with JEDEC standard JESD251 (profiles 1.0 and 2.0), JESD251-1 and JESD252.

JESD251 specifies two interface profiles where profile 1.0 is Octal SPI and profile 2.0 is HyperBus™ (HyperRAM™ and HyperFlash™). OSPI supports the QSPI protocol.

[Table 44.1](#) lists the OSPI specifications, [Figure 44.1](#) shows a block diagram, and [Table 44.2](#) lists the I/O pins.

Table 44.1 OSPI specifications

Item	Description
Number of units	2 units
Protocol	Compliant with the xSPI protocol*1
Data transmission and reception	Issue transaction for up to 2 slaves as masters. Only one of the memory devices can operate at a time.
Transfer speed	Support the transfer at xSPI333
Mode	- Support the following protocol modes: 1-, 4-, 8-pin with SDR/DDR (1S-1S-1S, 4S-4D-4D, 8D-8D-8D) 2-, 4-pin with SDR (1S-2S-2S, 2S-2S-2S, 1S-4S-4S, 4S-4S-4S) - Configurable address length - Configurable initial access latency cycle - Support XiP mode
OSPI function	- Support write data mask - Support in-band reset - Memory-mapping - Unit 0 supports up to 256 MB address space each CS - Unit 1 supports up to 128 MB address space each CS 1 control channel per unit for the bus masters For more details, see section 15, Buses. - Prefetch function for burst-read with low latency - Outstanding buffer for burst-write with high throughput - Manual command - Configurable up to 4 commands - Status register polling function - Input strobe port timing shift
Decryption function	Decryption on-the-fly is available for memory-map read
Interrupt source	- Error interrupt - Completion interrupt
Module-stop function	Module-stop state can be set to reduce power consumption
TrustZone Filter	- Security attribution can be set for the IO register area - External address space is defined as non-secure

Note 1. The OSPI is compliant with JEDEC standard JESD251 (profiles 1.0 and 2.0), JESD251-1 and JESD252.

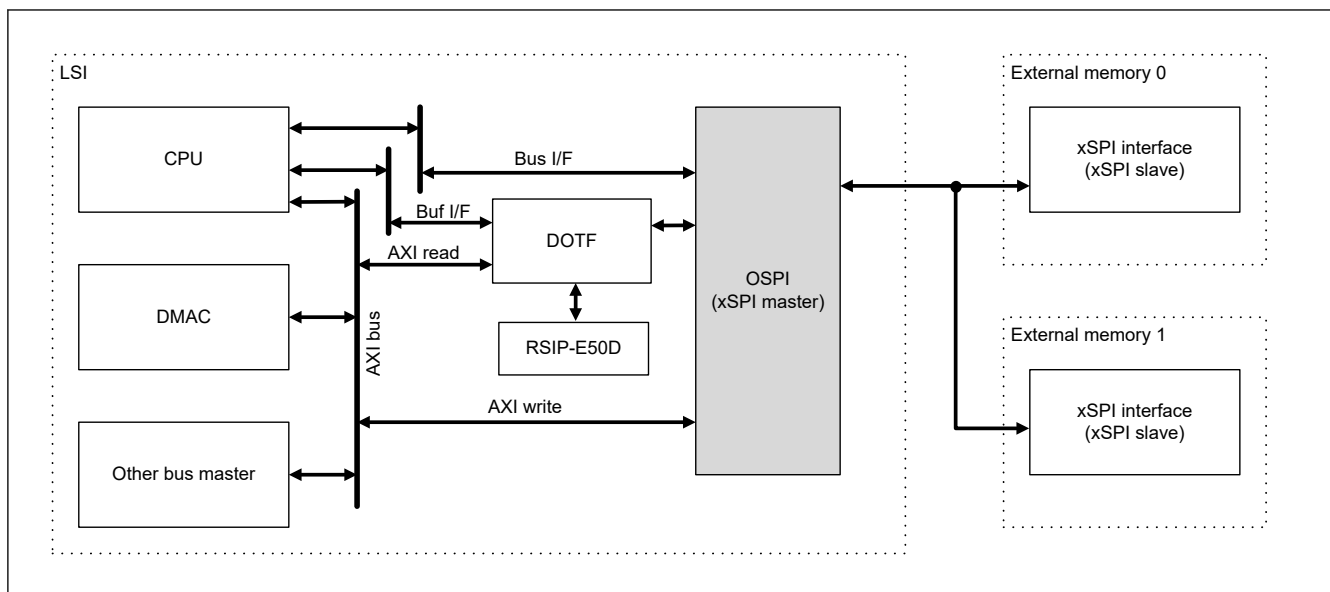


Figure 44.1 Block diagram

Table 44.2 OSPI I/O pins

Pin name	I/O	Function
OM_n_SCLK	Output	Clock positive
OM_n_SCLKN	Output	Clock negative
OM_n_CS0	Output	Chip selection for slave 0
OM_n_CS1	Output	Chip selection for slave 1
OM_n_DQS	I/O	Read data strobe/write data mask
OM_n_SIO0	I/O	Data 0 input/output
OM_n_SIO1	I/O	Data 1 input/output
OM_n_SIO2	I/O	Data 2 input/output
OM_n_SIO3	I/O	Data 3 input/output
OM_n_SIO4	I/O	Data 4 input/output
OM_n_SIO5	I/O	Data 5 input/output
OM_n_SIO6	I/O	Data 6 input/output
OM_n_SIO7	I/O	Data 7 input/output
OM_n_RESET	Output	Master reset status for slave 0, 1
OM_n_RSTO1	Input	Slave reset status for slave 1
OM_n_ECSINT1	Input	Interrupt for slave 1/error correction status for slave 1
OM_n_WP1	Output	Write protect for slave 1

Note: For OM_n_SIO7-0, OM_n_SCLK, OM_n_SCLKN and OM_n_DQS pins, 30Ω ±1% resistor must be put on the board to comply with JESD251 I/O driver definition. Renesas recommends this for the operation with proper signal quality.

Note: n = 0, 1

44.2 Register Descriptions

44.2.1 OSPI Configuration Registers

These registers configure the xSPI Master function. These registers should be configured in the initialization phase before issuing xSPI transactions.

44.2.1.1 WRAPCFG : OSPI Wrapper Configuration Register

Base address: $\text{OSPI}_{n_B} = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $\text{OSPI}_{n_B_NS} = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x000

Bit position:	31	28	24	20	16	12	8	4	0																	
Bit field:	—	—	—	DSSFTCS1[4:0]				—	—	—	—	—	—	—	DSSFTCS0[4:0]				—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
12:8	DSSFTCS0[4:0]	DS Shift for Slave 0 These bits configure the number of delay cells for an OM_DQS port. A delay cell is used to adjust the DS sampling timing. When automatic calibration is enabled, it can be updated automatically. In this case, it should not be written by the user. 0x00: No shift 0x01: Add a delay of one cell ⋮ 0x1E: Add a delay of 30 cells 0x1F: Add a delay of 31 cells	R/W
23:13	—	These bits are read as 0. The write value should be 0.	R/W
28:24	DSSFTCS1[4:0]	DS Shift for Slave 1 The function is the same as one of slave 0.	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The WRAPCFG register has functions to configure the xSPI Master function.

44.2.1.2 COMCFG : OSPI Common Configuration Register

Base address: $\text{OSPI}_{n_B} = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $\text{OSPI}_{n_B_NS} = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x004

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	OENEGEX	OEASTEX
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0. The write value should be 0.	R/W
16	OEASTEX	Output Enable Asserting Extension This bit extends one cycle output enable of data and DS during output enable asserting. When set to 1, CS asserting should be extended ($\text{LIOCFGCSn.CSASTEX} = 1$ ($n = 0, 1$)). This bit is used for case with latency cycle because OSPI output data can be conflicted with OSPI input data. 0: Do not extend one cycle output enable 1: Extend one cycle output enable	R/W

Bit	Symbol	Function	R/W
17	OENEGEX	Output Enable Negating Extension This bit extends one cycle output enable of data and DS during output enable negating. This bit is used for case with latency cycle because OSPI output data can be conflicted with OSPI input data. 0: Do not extend one cycle output enable 1: Extend one cycle output enable	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The COMCFG register has functions to configure the xSPI Master function.

44.2.1.3 BMCFGCHn : OSPI Bridge Map Configuration Register Chn (n = 0)

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ (n = 0, 1)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ (n = 0, 1)

Offset address: $0x008 + 0x004 \times n$

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	CMBTIM[7:0]								—	—	—	—	—	—	—	PREEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MWRSIZE[7:0]								MWR COMB	—	—	—	—	—	—	WRMD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	WRMD	System Bus Write Response Mode This bit selects the timing of a system bus write response in memory-mapping mode. When set to 1, this bit returns the response after transmitting a frame on the xSPI bus. When the system bus write response mode is enabled, the memory write combination mode must be disabled. 0: Return response after storing to the internal write buffer 1: Return response after issuing a write transaction to the xSPI bus	R/W
6:1	—	These bits are read as 0. The write value should be 0.	R/W
7	MWRCOMB	Memory Write Combination Mode This bit selects to combine the OSPI data in write access of memory-mapping mode. When set to 0, the OSPI data size depends on the burst type and size of the system bus. When this field is set to 1, the data size depends on the MWRSIZE[7:0] field.*1 When set to 1, any write transaction can be temporarily held in this xSPI master. 0: Disable combination mode 1: Enable combination mode	R/W
15:8	MWRSIZE[7:0]	Memory Write Size These bits select the size to combine incremental address in memory-mapping mode. It transmits an xSPI frame with the data combined up to the configured size while the address is incremental. When a nonincremental address or a read transaction is detected before reaching the target size, it transmits the pending data into the xSPI bus. 0x00: Combine incremental address of up to 4 bytes 0x01: Combine incremental address of up to 8 bytes ⋮ 0x0E: Combine incremental address of up to 60 bytes 0x0F: Combine incremental address of up to 64 bytes 0xFF: Combine incremental address of up to 2 bytes Others: Setting prohibited	R/W
16	PREEN	Prefetch Enable This bit enables the prefetch function for a read transaction in memory-mapping mode. It can reduce the latency for read transaction with incremental address. 0: Disable prefetch function 1: Enable prefetch function	R/W

Bit	Symbol	Function	R/W
23:17	—	These bits are read as 0. The write value should be 0.	R/W
31:24	CMBTIM[7:0]	Combination Timer These bits specify the expiration period of a combination timer. This timer is counted by PCLKA. 0x00 means disabling the combination timer. When the timer is expired, the data in the combination buffer is pushed to the memory device.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Writing to OSPI memory space other than 64 bit is prohibited during combination mode.

The BMCFGCHn register has functions to configure the xSPI Master function.

44.2.1.4 CMCFG0CSn : OSPI Command Map Configuration Register 0 CSn (n = 0, 1)

Base address: $OSPI_{n_B} = 0x4026_8000 + 0x0400 \times n$ (n = 0, 1)
 $OSPI_{n_B_NS} = 0x5026_8000 + 0x0400 \times n$ (n = 0, 1)

Offset address: $0x010 + 0x010 \times n$

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	ADDRPCD[7:0]								ADDRPEN[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	ARYA MD	WPBS TMD	ADDSIZE[1:0]	FFMT[1:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	FFMT[1:0]	Frame Format These bits configure xSPI frame format in memory-mapping mode. See Table 44.7 for detail. 0 0: Normal format: Command 1 byte, address ADDSIZE, data up to system bus transaction 0 1: 8D-8D-8D profile 1.0 format: Command 2 bytes, address ADDSIZE, data up to system bus transaction 1 0: 8D-8D-8D profile 2.0 Command Modifier format: Command & Modifier 6 bytes, data up to system bus transaction 1 1: 8D-8D-8D profile 2.0 Commands with Extended Command Modifier format: Command & Modifier 6 bytes, data up to system bus transaction	R/W
3:2	ADDSIZE[1:0]	Address Size These bits configure the number of address byte in memory-mapping mode. For 8D-8D-8D profile 2.0, configure to 4 bytes. 0 0: 1 byte (256-byte address space) 0 1: 2 bytes (64-KB address space) 1 0: 3 bytes (16-MB address space) 1 1: 4 bytes (4-GB address space)	R/W
4	WPBSTMD	Wrapping Burst Mode When this bit is set to 1, the wrapping boundary between system bus access and xSPI memory should be matched. 0: Separate xSPI transfer at the wrapping address boundary 1: Do not separate xSPI transfer at the wrapping address boundary	R/W
5	ARYAMD	Array Address Mode When this bit is set to 1, address for memory is mapped as {A[25:10], A[9:4], 6{RSV}, A[3:0]} where A[25:0] is normal address, and RSV is reserved value(0b). This bit is valid only when FFMT = 1d. 0: Normal address mode 1: Array address mode	R/W
15:6	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W											
23:16	ADDRPEN[7:0]	Address Replace Enable These bits select the bits to replace the MSByte of system bus address in memory-mapping mode. 0: No replacement (xSPI frame address field is the same as system bus address) 1: Replacement	R/W											
		<table><tr><th>ADDSSIZE[1:0]</th><th>Replace target bits of system bus address</th></tr><tr><td>00</td><td>bit 7 to bit 0</td></tr><tr><td>01</td><td>bit 15 to bit 8</td></tr><tr><td>10</td><td>bit 23 to bit 16</td></tr><tr><td>11</td><td>bit 31 to bit 24</td></tr></table>		ADDSSIZE[1:0]	Replace target bits of system bus address	00	bit 7 to bit 0	01	bit 15 to bit 8	10	bit 23 to bit 16	11	bit 31 to bit 24	
		ADDSSIZE[1:0]		Replace target bits of system bus address										
		00		bit 7 to bit 0										
		01		bit 15 to bit 8										
		10		bit 23 to bit 16										
11	bit 31 to bit 24													
31:24	ADDRPCD[7:0]	Address Replace Code These bits configure the code to replace the MSByte of system bus address in memory-mapping mode. It replaces the corresponding bits when the Address Replace Enable bit is set to 1.	R/W											

Note: S-TYPE-3, P-TYPE-3

The CMCFG0CSn register has functions to configure the xSPI Master function.

44.2.1.5 CMCFG1CSn : OSPI Command Map Configuration Register 1 CSn (n = 0, 1)

Base address: $OSPI_{n_B} = 0x4026_8000 + 0x0400 \times n$ (n = 0, 1)
 $OSPI_{n_B_NS} = 0x5026_8000 + 0x0400 \times n$ (n = 0, 1)

Offset address: $0x014 + 0x010 \times n$

Bit position:	31	20	16	15	0
Bit field:	—	—	—	—	—
Value after reset:	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	RDCMD[15:0]	Read Command These bits configure the command field of a read transaction in memory-mapping mode. Normal format and 8D-8D-8D profile 2.0 format use only upper 1 byte. 8D-8D-8D profile 1.0 format uses 2 bytes.	R/W
20:16	RDLATE[4:0]	Read Latency Cycle These bits configure the latency cycle of a read transaction in memory-mapping mode. 0x00: No latency 0x01: 1 cycle : : 0x08: 8 cycles (default) : : 0x1E: 30 cycles 0x1F: 31 cycles Others: Setting prohibited	R/W
31:21	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The CMCFG1CSn register has functions to configure the xSPI Master function.

44.2.1.6 CMCFG2CSn : OSPI Command Map Configuration Register 2 CSn (n = 0, 1)

Base address: $OSPI_{n_B} = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $OSPI_{n_B_NS} = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: $0x018 + 0x010 \times n$

Bit position: 31 20 16 15 0

Bit field:	—	—	—	—	—	—	—	—	—	—	WRLATE[4:0]	WRCMD[15:0]
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Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	WRCMD[15:0]	Write Command These bits configure the command field of a write transaction in memory-mapping mode. Normal format and 8D-8D-8D profile 2.0 format use only upper 1 byte. 8D-8D-8D profile 1.0 format uses 2 bytes.	R/W
20:16	WRLATE[4:0]	Write Latency Cycle These bits configure the latency cycle of a write transaction in memory-mapping mode. 0x00: No latency 0x01: 1 cycle ⋮ ⋮ 0x08: 8 cycles (default) ⋮ ⋮ 0x1E: 30 cycles 0x1F: 31 cycles Others: Setting prohibited	R/W
31:21	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The CMCFG2CSn register has functions to configure xSPI the Master function.

44.2.1.7 LIOCFGCSn : OSPI Link I/O Configuration Register CSn (n = 0, 1)

Base address: $OSPln_B = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $OSPln_B\ NS = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: $0x050 + 0x004 \times n$

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field:	DDSRMPEX[3:0]	SDSRMPSFT[3:0]	SDSRMPMD	SDSRD RV	CSNE GEX	CSAS TEX	CSMIN[3:0]
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Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:	—	—	—	—	WRMS KMD	LATE MD	PRTMD[9:0]
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Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
9:0	PRTMD[9:0]	Protocol Mode These bits configure the protocol mode and the pin to sample data inputs. When using data strobe instead of SPI clock for sampling in SDR mode, set PRTMD[9:0] to 1. 0x000: 1S-1S-1S 0x3B2: 4S-4D-4D 0x3FF: 8D-8D-8D 0x048: 1S-2S-2S 0x049: 2S-2S-2S 0x090: 1S-4S-4S 0x092: 4S-4S-4S Others: Setting prohibited	R/W

Bit	Symbol	Function	R/W													
10	LATEMD	Latency Mode This bit selects the behavior of the initial access latency phase for both direct-manual mode and memory-mapping mode. When set to 0, the latency cycle is equal to each configured cycle from the transmitting Address field. When set to 1, the latency cycle is incremented from the last byte pair of the Address field and is extended 2 times of each configured cycle depending on the OM_DQS port. Latency mode is used only for profile 2.0 frame format of 8D-8D-8D protocol mode with 6 bytes Command/Address field. See the xSPI protocol specification for detail. <table><tr><th>Value: Function</th><th>Frame format</th><th>Usage</th></tr><tr><td rowspan="2">0: Configurable latency</td><td>Profile 2.0</td><td>The configurable latency cycle should be set as minus 1.</td></tr><tr><td>Others</td><td>The latency cycle increments after the Address field.</td></tr><tr><td rowspan="2">1: Variable latency</td><td>Profile 2.0</td><td>The latency cycle increments from address [23:16] and should not be set to 1.</td></tr><tr><td>Others</td><td>Not support</td></tr></table>	Value: Function	Frame format	Usage	0: Configurable latency	Profile 2.0	The configurable latency cycle should be set as minus 1.	Others	The latency cycle increments after the Address field.	1: Variable latency	Profile 2.0	The latency cycle increments from address [23:16] and should not be set to 1.	Others	Not support	R/W
Value: Function	Frame format	Usage														
0: Configurable latency	Profile 2.0	The configurable latency cycle should be set as minus 1.														
	Others	The latency cycle increments after the Address field.														
1: Variable latency	Profile 2.0	The latency cycle increments from address [23:16] and should not be set to 1.														
	Others	Not support														
11	WRMSKMD	Write Mask Mode This bit selects to use the OM_DQS port as the write data mask. It can be useful for write access of odd byte. Write mask mode is used only for 8D-8D-8D protocol mode. 0: Write mask disable 1: Write mask enable	R/W													
15:12	—	These bits are read as 0. The write value should be 0.	R/W													
19:16	CSMIN[3:0]	CS Minimum Idle Term This bit configures the minimum cycle between xSPI frames. 0x0: 1 cycle 0x1: 2 cycles ⋮ 0x7: 8 cycles (default) ⋮ 0xE: 15 cycles 0xF: 16 cycles Others: Setting prohibited	R/W													
20	CSASTEX	CS Asserting Extension This bit extends 1 cycle Chip Select pins when asserting. 0: No extension 1: Extend 1 cycle	R/W													
21	CSNEGEX	CS Negating Extension This bit extends 1 cycle Chip Select pins when negating. 0: No extension 1: Extend 1 cycle	R/W													
22	SDRDRV	SDR Driving Timing This bit configures the timing of data output in SDR. This bit should not be set to 1 when there is no latency cycle because OSPI output data can be conflicted with OSPI input data. 0: Drive at 1/2 cycle before CK rising edge 1: Drive at CK rising-edge	R/W													
23	SDRSMPMD	SDR Sampling Mode This bit selects the edge of sampling in SDR. In DDR, regardless of the setting, this bit samples OM_SIO_n (n = 0 to 7) ports with both edges of OM_DQS. When set to 1, this bit samples at rising edge before falling edge. 0: Samples data input at falling edge 1: Samples data input at rising edge	R/W													

Bit	Symbol	Function	R/W
27:24	SDRSMPST[3:0]	SDR Sampling window shift These bits shift the timing of CK sampling in SDR. When using OM_DQS in SDR, there is no influence on the behavior. For DDR or when using OM_DQS in SDR, set these bits to 0. 0x0: Sample without delay 0x1: Sample at 1-cycle delay : : 0x6: Sample at 6-cycle delay 0x7: Sample at 7-cycle delay Others: Setting prohibited	R/W
31:28	DDRSMPPEX[3:0]	DDR Sampling Window Extend These bits configure the cycle of extending the sampling window in DDR. In DDR, the input data is sampled during the expected cycle soon after the latency cycle. The input data out of range is ignored. It can be configured depending on the OM_DQS propagation delay. These bits setting should be set to satisfy t_{CKHDSH} in section 60.3.11. OSPI Timing. t_{CKHDSH} means the output delay time of the external memory. For example, the DDRSMPEX[3:0] bits are set to 0x1 in the following conditions. - VCC is 2.7 V or above - OM_n_SCLK is 133 MHz - Memory output delay time (t_{CKHDSH}) is 6 ns The maximum allowable t_{CKHDSH} value is: $t_{PERIOD} \times (1 + \text{DDRSMPPEX}[3:0]) - 8.5 = 7.5 \times (1 + 1) - 8.5 = 6.5[\text{ns}]$ The memory output delay time should be less than this value. 0x0: Expand no cycle 0x1: Expand 1 cycle : : 0x6: Expand 6 cycles 0x7: Expand 7 cycles Others: Setting prohibited	R/W

Note: S-TYPE-3, P-TYPE-3

The LIOCFGCSn register has functions to configure the xSPI Master function.

44.2.2 OSPI Control Registers

These registers control the xSPI Master function.

44.2.2.1 BMCTL0 : OSPI Bridge Map Control Register 0

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x060

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	CH0CS1ACC [1:0]	CH0CS0ACC [1:0]	CH0CS0ACC [1:0]	CH0CS0ACC [1:0]
Value after reset:	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
1:0	CH0CS0ACC[1:0]	System bus ch0 to slave 0 memory area access enable This field enables the access from ch0 to CS0 memory. 0 0: Read/write disable 0 1: Read enable, write disable 1 0: Read disable, write enable 1 1: Read/write enable	R/W

Bit	Symbol	Function	R/W
3:2	CH0CS1ACC[1:0]	System bus ch0 to slave 1 memory area access enable This field enables the access from ch0 to CS1 memory. 0 0: Read/write disable 0 1: Read enable, write disable 1 0: Read disable, write enable 1 1: Read/write enable	R/W
31:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The BMCTL0 register has functions to control the xSPI Master function.

This register is configured in the initialization phase. When the setting requires modifications after the start of an xSPI transaction, stop all communication. See [section 44.3.7.2. Flow of Communication Stop](#) before changing the value of BMCTL0.

44.2.2.2 BMCTL1 : OSPI Bridge Map Control Register 1

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x064

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	PBUF CLRC H0	—	MWRP USHC H0	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	The write value should be 0.	W
8	MWRPUSHCH0	Memory Write Data Push for ch0 This field requests to push the pending data in combination mode.*1 0: No command 1: Push request	W
9	—	The write value should be 0.	W
10	PBUFCLRCH0	Prefetch Buffer Clear for ch0 This field requests to clear the prefetch buffer when the prefetch function is enabled.*1 Do not set this bit during memory access (COMSTT.MEMACCCH0 = 1). 0: No command 1: Clear request	W
31:11	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

Note 1. Prefetch/Combination behavior is not defined when the cycle base race condition between asserting these bits and system bus access has occurred.

The BMCTL1 register has functions to control the xSPI Master function.

44.2.2.3 CMCTLCHn : OSPI Command Map Control Register CHn (n = 0)

Base address: $\text{OSPI}_{n_B} = 0x4026_8000 + 0x0400 \times n$ (n = 0, 1)
 $\text{OSPI}_{n_B_NS} = 0x5026_8000 + 0x0400 \times n$ (n = 0, 1)

Offset address: $0x0068 + 0x0004 \times n$

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	XIPEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	XIPEXCODE[7:0]								XIPENCODE[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	XIPENCODE[7:0]	XiP Mode Enter Code These bits configure the code to enter XiP mode in memory-mapping mode.	R/W
15:8	XIPEXCODE[7:0]	XiP Mode Exit Code These bits configure the code to exit XiP mode in memory-mapping mode.	R/W
16	XIPEN	XiP Mode Enable This bit enables XiP mode in memory-mapping mode. When set to 1, XiP enter code is inserted in the latency field, and the command field in the next transaction is omitted. When set to 0, XiP exit code is inserted in the latency field. This bit is set to 0 automatically when transmitting a XiP disable pattern. XiP mode should not be used for 8D-8D-8D protocol mode profile 2.0 frame format. 0: Disable XiP mode 1: Enable XiP mode	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The CMCTLCHn register has functions to control the xSPI Master function.

44.2.2.4 CDCTL0 : OSPI Command Manual Control Register 0

Base address: $\text{OSPI}_{n_B} = 0x4026_8000 + 0x0400 \times n$ (n = 0, 1)
 $\text{OSPI}_{n_B_NS} = 0x5026_8000 + 0x0400 \times n$ (n = 0, 1)

Offset address: 0x070

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	PERREP[3:0]				—	—	—	PERITV[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	TRNUM[1:0]	CSSE L	—	PERM D	TRRE Q	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TRREQ	Transaction Request This bit requests to issue the transaction of manual-command. When set to 1, it starts the transaction. This bit is cleared to 0 when the transaction is completed. The transaction is canceled by clearing to 0 while the transaction is ongoing. 0: No transaction 1: Request transaction	R/W

Bit	Symbol	Function	R/W
1	PERMD	Periodic Mode This bit enables the periodic transaction mode. When set to 1, it repeats a transaction periodically and compares the read value with the expected value. Periodic transaction mode alternates the status polling operation for external memory. 0: Direct manual-command mode 1: Periodic manual-command mode	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
3	CSSEL	Chip Select This bit selects a target memory to issue a manual-command. 0: CS0 1: CS1	R/W
5:4	TRNUM[1:0]	Transaction Number These bits configure the number of transactions in normal manual-command mode. In periodic manual-command, regardless of this setting, the read data of the last command is compared. 0 0: Issue 1 command (using command buffer 0) 0 1: Issue 2 commands (using command buffer 0-1) 1 0: Issue 3 commands (using command buffer 0-2) 1 1: Issue 4 commands (using command buffer 0-3)	R/W
15:6	—	These bits are read as 0. The write value should be 0.	R/W
20:16	PERITV[4:0]	Periodic Transaction Interval These bits configure the interval of transaction in periodic manual-command mode. Too short interval compared with CPU bus cycle can result in no store into command buffer0. The interval should be longer than 4 times the CPU bus cycle. 0x00: 2 (= 2 ¹) cycles 0x01: 4 (= 2 ²) cycles ⋮ 0x1E: 2,147,483,648 (= 2 ³¹) cycles 0x1F: 4,294,967,296 (= 2 ³²) cycles	R/W
23:21	—	These bits are read as 0. The write value should be 0.	R/W
27:24	PERREP[3:0]	Periodic Transaction Repeat These bits configure the number of transaction repetitions in periodic manual-command mode. 0x0: 1 (= 2 ⁰) time 0x1: 2 (= 2 ¹) times ⋮ 0xE: 16384 (= 2 ¹⁴) times 0xF: 32768 (= 2 ¹⁵) times	R/W
31:28	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The CDCTL0 register has functions to control the xSPI Master function.

44.2.2.5 CDCTL1 : OSPI Command Manual Control Register 1

Base address: OSPI_n_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPI_n_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x074

Bit position: 31

0

Bit field:

PEREXP[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	PEREXP[31:0]	Periodic Transaction Expected Value These bits configure the expected value to compare with the read value in periodic manual-command mode. For example, when comparing one byte, the lower byte should be configured.	R/W

Note: S-TYPE-3, P-TYPE-3

The CDCTL1 register has functions to control the xSPI Master function.

44.2.2.6 CDCTL2 : OSPI Command Manual Control Register 2

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x078

Bit position: 31

0

Bit field:

PERMSK[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	PERMSK[31:0]	Periodic Transaction Masked Value These bits configure the masked value for the expected value in periodic manual-command mode. When 1 is set to any bit, the corresponding bit is configured as the expected value (CDCTL1.PEREXP[31:0]) is ignored. In 8D-8D-8D, the data bytes are transferred only in byte pairs on the xSPI bus. This means the dummy read data can be stored. It should be masked for unused bits. For example, when reading a lower one byte, it should be configured to 0xFFFFF00.	R/W

Note: S-TYPE-3, P-TYPE-3

The CDCTL2 register has functions to control the xSPI Master function.

44.2.2.7 CDTBUFn : OSPI Command Manual Type Buf n ($n = 0$ to 3)

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: $0x080 + 0x010 \times n$

Bit position:

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field:

CMD[15:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position:

15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:

TRTY
PE

—

LATE[4:0]

DATASIZE[3:0]

ADDSSIZE[2:0]

CMDSIZE[1:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
1:0	CMDSIZE[1:0]	Command Size These bits configure the size of the Command field. In 8D-8D-8D, set these bits to 10b. Do not configure both command size and address size to 0. 0 0: 0 bytes (no command phase) 0 1: 1 byte 1 0: 2 bytes Others: Setting prohibited	R/W

Bit	Symbol	Function	R/W
4:2	ADDSSIZE[2:0]	Address Size These bits configure the size of the address field. 0 0 0: 0 bytes (no address phase) 0 0 1: 1 byte 0 1 0: 2 bytes 0 1 1: 3 bytes 1 0 0: 4 bytes Others: Setting prohibited	R/W
8:5	DATASIZE[3:0]	Write/Read Data Size These bits configure the size of the Data field. In 8D-8D-8D, the data bytes are transferred only in byte pairs on the xSPI bus. For example, even when configuring 1 byte for read, 2 bytes of data are received. The last byte should be ignored. Do not configure 0 bytes for read transaction. 0x0: 0 bytes (no data phase) 0x1: 1 byte : : 0x7: 7 bytes 0x8: 8 bytes Others: Setting prohibited	R/W
13:9	LATE[4:0]	Latency Cycle These bits configure the latency cycle in manual-command mode. 0x0: No latency 0x1: 1 cycle : : 0x1E: 30 cycles 0x1F: 31 cycles	R/W
14	—	This bit is read as 0. The write value should be 0.	R/W
15	TRTYPE	Transaction Type This bit selects the type of transaction. 0: Read transaction (readout data from slave device) 1: Not read transaction	R/W
31:16	CMD[15:0]	Command (1-2 bytes) These bits configure the Command field in manual-command mode. The number of bytes configured in Command Size bit is transferred. 1S-1S-1S, 4S-4D-4D: CMD[15:8] is Command field, CMD[7:0] is not used. 8D-8D-8D profile 1.0: CMD[15:8] is Command field, CMD[7:0] is Extension field. 8D-8D-8D profile 2.0: CMD[15:0] is the upper 2 bytes of the Command & Modifier fields (bits [47:32] in the xSPI protocol)	R/W

Note: S-TYPE-3, P-TYPE-3

The CDTBUF_n register has functions to control the xSPI Master function.

44.2.2.8 CDABUF_n : OSPI Command Manual Address Buf n (n = 0 to 3)

Base address: $OSPI_{n_B} = 0x4026_8000 + 0x0400 \times n$ (n = 0, 1)
 $OSPI_{n_B_NS} = 0x5026_8000 + 0x0400 \times n$ (n = 0, 1)

Offset address: $0x084 + 0x10 \times n$

Bit position: 31

0

Bit field:

ADD[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	ADD[31:0]	Address These bits configure the Address field in manual-command mode. 1S-1S-1S, 4S-4D-4D, 8D-8D-8D profile 1.0: The Address field. 8D-8D-8D profile 2.0: The lower 4 bytes of the Command & Modifier fields (bits [31:0] in the xSPI protocol).	R/W

Note: S-TYPE-3, P-TYPE-3

The CDABUF_n register has functions to control the xSPI Master function.

44.2.2.9 CDD0BUF_n : OSPI Command Manual Data 0 Buf _n (n = 0 to 3)

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ (n = 0, 1)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ (n = 0, 1)

Offset address: $0x088 + 0x10 \times n$

Bit position: 31

0

Bit field:

DATA[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	DATA[31:0]	Write/Read Data These bits configure the Data field in manual-command mode. For write transaction, configure the write data. For read transaction, the read data is stored.	R/W

Note: S-TYPE-3, P-TYPE-3

The CDD0BUF_n register has functions to control the xSPI Master function.

44.2.2.10 CDD1BUF_n : OSPI Command Manual Data 1 Buf _n (n = 0 to 3)

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ (n = 0, 1)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ (n = 0, 1)

Offset address: $0x08C + 0x10 \times n$

Bit position: 31

0

Bit field:

DATA[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	DATA[31:0]	Write/Read Data These bits configure the Data field in manual-command mode. For write transaction, configure the write data. For read transaction, the read data is stored.	R/W

Note: S-TYPE-3, P-TYPE-3

The CDD1BUF_n register has functions to control the xSPI Master function.

44.2.2.11 LPCTL0 : OSPI Link Pattern Control Register 0

Base address: $\text{OSPI}_{n_B} = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $\text{OSPI}_{n_B_NS} = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x100

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	XD2VAL	—	—	XD2LEN[4:0]				XD1VAL	—	—	XD1LEN[4:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	XD1PIN[1:0]	CSSEL	—	—	PATREQ	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PATREQ	Pattern Request This bit requests to issue the pattern. When set to 1, it starts the pattern. It is cleared to 0 when the pattern is completed. 0: No request XiP disable pattern 1: Request XiP disable pattern	R/W
2:1	—	These bits are read as 0. The write value should be 0.	R/W
3	CSSEL	Chip Select This bit selects a target memory to issue a pattern. 0: Slave 0 (CS0) 1: Slave 1 (CS1)	R/W
5:4	XD1PIN[1:0]	XiP Disable Pattern Pin These bits select the data output pins to transmit XiP disable pattern. 0 0: 1 pin 0 1: 2 pins 1 0: 4 pins 1 1: 8 pins	R/W
15:6	—	These bits are read as 0. The write value should be 0.	R/W
20:16	XD1LEN[4:0]	XiP Disable Pattern 1st Phase Length These bits select the length of 1st phase in XiP disable pattern. Do not configure the pattern with zero-length for both 1st phase and 2nd phase. 0x0: 0 cycles 0x1: 1 cycle ⋮ 0x1E: 30 cycles 0x1F: 31 cycles	R/W
22:21	—	These bits are read as 0. The write value should be 0.	R/W
23	XD1VAL	XiP Disable Pattern 1st Phase Value This bit selects the value of 1st phase in XiP disable pattern. 0: Low drive 1: High drive	R/W
28:24	XD2LEN[4:0]	XiP Disable Pattern 2nd Phase Length These bits select the length of 2nd phase in XiP disable pattern. 0x00: 0 cycles 0x01: 1 cycle ⋮ 0x1E: 30 cycles 0x1F: 31 cycles	R/W
30:29	—	These bits are read as 0. The write value should be 0.	R/W
31	XD2VAL	XiP Disable Pattern 2nd Phase Value This bit selects the value of 2nd phase in XiP disable pattern. 0: Low drive 1: High drive	R/W

Note: S-TYPE-3, P-TYPE-3

The LPCTL0 register has functions to control the xSPI Master function.

44.2.2.12 LPCTL1 : OSPI Link Pattern Control Register 1

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x104

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	RSTSU[2:0]			—	RSTWID[2:0]			—	—	RSTREP[1:0]		CSSEL	—	PATREQ[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	PATREQ[1:0]	Pattern Request These bits request to issue the pattern. When set to 01b or 10b, it starts the pattern. It is cleared to 00b when the pattern is completed. 0 0: No request 0 1: Request Reset pattern 1 0: Request CS-only pattern 1 1: Setting prohibited	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
3	CSSEL	Chip select This bit selects a target memory to issue a pattern. 0: Slave 0 (CS0) 1: Slave 1 (CS1)	R/W
5:4	RSTREP[1:0]	Reset Pattern Repeat These bits select the repeating time to toggle CS from low to high. 0 0: 4 times (specified on Reset Signaling Protocol) 0 1: 5 times 1 0: 6 times 1 1: 7 times	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
10:8	RSTWID[2:0]	Reset Pattern Width These bits configure the width of a cycle in Reset pattern and CS-only pattern. It toggles CS with the configured cycle. 0 0 0: 2 (= 2 ¹) cycles 0 0 1: 4 (= 2 ²) cycles ⋮ 1 1 0: 128 (= 2 ⁷) cycles 1 1 1: 256 (= 2 ⁸) cycles	R/W
11	—	This bit is read as 0. The write value should be 0.	R/W
14:12	RSTSU[2:0]	Reset Pattern Data Output Setup Time These bits configure the number of setup cycles for data output based on the edge of CS in a Reset pattern. The number of setup cycles requires enough setup time because the xSPI slave samples any data at the rising edge of CS. This cycle of setup time should be less than the cycle of reset pattern width (RSTWID[2:0]). 0 0 0: 1 cycle 0 0 1: 2 cycles ⋮ 1 1 0: 7 cycles 1 1 1: 8 cycles	R/W
31:15	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The LPCTL1 register has functions to control the xSPI Master function.

44.2.2.13 LIOCTL : OSPI Link I/O Control Register

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x108

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RSTC S0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	WPCS 1	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

Bit	Symbol	Function	R/W
0	—	This bit is read as 1. The write value should be 1.	R/W
1	WPCS1	WP Drive for Slave 1 This bit controls the value of the OM_WP1 port. It can be useful only for xSPI slave with write protect port. 0: Drive low level 1: Drive high level	R/W
15:2	—	These bits are read as 0. The write value should be 0.	R/W
16	RSTCS0	Reset Drive This bit controls the value of OM_RESET port. It can be useful only for xSPI slave with reset port. 0: Drive low level 1: Drive high level	R/W
17	—	This bit is read as 1. The write value should be 1.	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The LIOCTL register has functions to control the xSPI Master function.

44.2.2.14 CCCTL0CSn : OSPI Command Calibration Control Register 0 CSn ($n = 0, 1$)

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: $0x130 + 0x020 \times n$

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	CASFTEND[4:0]					—	—	—	CASFTSTA[4:0]				
Value after reset:	0	0	0	1	1	1	1	1	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	CAITV[4:0]					—	—	—	—	—	—	CANO WR	CAEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CAEN	Automatic Calibration Enable This bit enables the automatic calibration. When set to 1, it transmits the calibration sequence periodically and adjusts the value of phase shift. When set to 0 during the calibration sequence, it stops after completion of the ongoing calibration sequence, this bit is then cleared. 0: Disable automatic calibration 1: Enable automatic calibration	R/W
1	CANOWR	Calibration No Write Mode This bit selects to omit write command in the calibration sequence. It can be used for any slave device with fixed calibration pattern data. 0: Calibration sequence with write command 1: Calibration sequence without write command	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
12:8	CAITV[4:0]	Calibration Interval These bits configure the intervals between calibration patterns. 0x00: 2 (= 2 ¹) cycle wait 0x01: 4 (= 2 ²) cycle wait ⋮ 0x1E: 2,147,483,648 (= 2 ³¹) cycle wait 0x1F: 4,294,967,296 (= 2 ³²) cycle wait	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
20:16	CASFTSTA[4:0]	Calibration DS Shift Start Value These bits configure the start value of DS shift.	R/W
23:21	—	These bits are read as 0. The write value should be 0.	R/W
28:24	CASFTEND[4:0]	Calibration DS Shift End Value These bits configure the end value of DS shift. The end value should be equal to or more than the start value (CASFTSTA[4:0]).	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The CCCTL0CSn register has functions to control the xSPI Master function.

44.2.2.15 CCCTL1CSn : OSPI Command Calibration Control Register 1 CSn (n = 0, 1)

Base address: OSPI_n_B = 0x4026_8000 + 0x0400 × n (n = 0, 1)
OSPI_n_B_NS = 0x5026_8000 + 0x0400 × n (n = 0, 1)

Offset address: 0x134 + 0x020 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	CARDLATE[4:0]				—	—	—	CAWRATE[4:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CADATASIZE[3:0]				CAADDSIZE[2:0]		CACMDSIZE [1:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	CACMDSIZE[1:0]	Command Size These bits configure the size of the Command field. In 8D-8D-8D, set these bits to 10b. Do not configure both command size and address size to 0. 0 0: 0 bytes (no command phase) 0 1: 1 byte 1 0: 2 bytes 1 1: Setting prohibited	R/W

Bit	Symbol	Function	R/W
4:2	CAADDSIZE[2:0]	Address Size These bits configure the size of the Address field. 0 0 0: 0 bytes (no address phase) 0 0 1: 1 byte 0 1 0: 2 bytes 0 1 1: 3 bytes 1 0 0: 4 bytes Others: Setting prohibited	R/W
8:5	CADATASIZE[3:0]	Write/Read Data Size These bits configure the size of the Data field. In 8D-8D-8D, the data size should be configured with an even byte. 0x0: 1 byte 0x1: 2 bytes : : 0xE: 15 bytes 0xF: 16 bytes	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W
20:16	CAWRLATE[4:0]	Write Latency Cycle These bits configure the latency cycle in the calibration frame. 0x00: No latency 0x01: 1 cycle : : 0x1E: 30 cycles 0x1F: 31 cycles	R/W
23:21	—	These bits are read as 0. The write value should be 0.	R/W
28:24	CARDLATE[4:0]	Read Latency Cycle These bits configure the latency cycle in the calibration frame. 0x00: No latency 0x01: 1 cycle : : 0x1E: 30 cycles 0x1F: 31 cycles	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

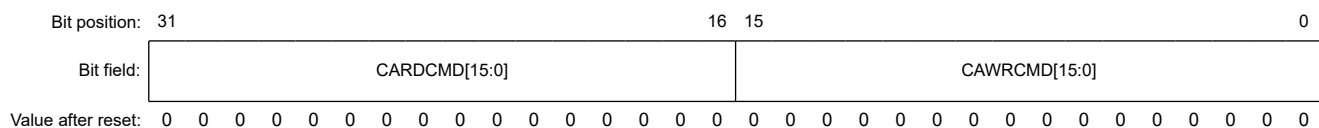
Note: S-TYPE-3, P-TYPE-3

The CCCTL1CSn register has functions to control the xSPI Master function. It should be updated while Automatic Calibration Enable is disabled.

44.2.2.16 CCCTL2CSn : OSPI Command Calibration Control Register 2 CSn (n = 0, 1)

Base address: $\text{OSPI}_{\text{In}}_{\text{B}} = 0x4026_8000 + 0x0400 \times n$ (n = 0, 1)
 $\text{OSPI}_{\text{In}}_{\text{B}}_{\text{NS}} = 0x5026_8000 + 0x0400 \times n$ (n = 0, 1)

Offset address: $0x138 + 0x020 \times n$



Bit	Symbol	Function	R/W
15:0	CAWRCMD[15:0]	Calibration Pattern Write Command These bits configure the calibration pattern write command.	R/W
31:16	CARDCMD[15:0]	Calibration Pattern Read Command These bits configure the calibration pattern read command.	R/W

Note: S-TYPE-3, P-TYPE-3

The CCCTL2CSn register has functions to control the xSPI Master function. It should be updated while Automatic Calibration Enable is disabled.

44.2.2.17 CCCTL3CSn : OSPI Command Calibration Control Register 3 CSn (n = 0, 1)

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ (n = 0, 1)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ (n = 0, 1)

Offset address: $0x13C + 0x020 \times n$

Bit position: 31

0

Bit field:

CAADD[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CAADD[31:0]	Calibration Pattern Address These bits configure the calibration pattern address.	R/W

Note: S-TYPE-3, P-TYPE-3

The CCCTL3CSn register has functions to control the xSPI Master function. It should be updated while Automatic Calibration Enable is disabled.

44.2.2.18 CCCTL4CSn : OSPI Command Calibration Control Register 4 CSn (n = 0, 1)

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ (n = 0, 1)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ (n = 0, 1)

Offset address: $0x140 + 0x020 \times n$

Bit position: 31

0

Bit field:

CADATA[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CADATA[31:0]	Calibration Pattern Data These bits configure the calibration pattern data.	R/W

Note: S-TYPE-3, P-TYPE-3

The CCCTL4CSn register has functions to control the xSPI Master function. It should be updated while Automatic Calibration Enable is disabled.

44.2.2.19 CCCTL5CSn : OSPI Command Calibration Control Register 5 CSn (n = 0, 1)

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ (n = 0, 1)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ (n = 0, 1)

Offset address: $0x144 + 0x020 \times n$

Bit position: 31

0

Bit field:

CADATA[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CADATA[31:0]	Calibration Pattern Data These bits configure the calibration pattern data.	R/W

Note: S-TYPE-3, P-TYPE-3

The CCCTL5CSn register has functions to control the xSPI Master function. It should be updated while Automatic Calibration Enable is disabled.

44.2.2.20 CCCTL6CSn : OSPI Command Calibration Control Register 6 CSn (n = 0, 1)

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ (n = 0, 1)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ (n = 0, 1)

Offset address: $0x148 + 0x020 \times n$

Bit position: 31

0

Bit field:

CADATA[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CADATA[31:0]	Calibration Pattern Data These bits configure the calibration pattern data.	R/W

Note: S-TYPE-3, P-TYPE-3

The CCCTL6CSn register has functions to control the xSPI Master function. It should be updated while Automatic Calibration Enable is disabled.

44.2.2.21 CCCTL7CSn : OSPI Command Calibration Control Register 7 CSn (n = 0, 1)

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ (n = 0, 1)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ (n = 0, 1)

Offset address: $0x14C + 0x020 \times n$

Bit position: 31

0

Bit field:

CADATA[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CADATA[31:0]	Calibration Pattern Data These bits configure the calibration pattern data.	R/W

Note: S-TYPE-3, P-TYPE-3

The CCCTL7CSn register has functions to control the xSPI Master function. It should be updated while Automatic Calibration Enable is disabled.

44.2.2.22 RACTL : OSPI Register Access Control Register

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ (n = 0, 1)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ (n = 0, 1)

Offset address: 0x208

Bit position: 31

30

29

28

27

26

25

24

23

22

21

20

19

18

17

16

Bit field:

—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position: 15

14

13

12

11

10

9

8

7

6

5

4

3

2

1

0

Bit field:

—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RAEN
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	------

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	RAEN	Register Access Enable 0: Disable MER register access 1: Enable MER register access	R/W

Bit	Symbol	Function	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

44.2.2.23 MER : OSPI Map End Replace Register

Base address: $OSPI_{n_B} = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $OSPI_{n_B_NS} = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x228

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	MECS1								—	—	—	MERC S1
Value after reset:	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
11:4	—	These bits are read as 1. The write value should be 1.	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W
16	MERCS1	Map End Replace CS1 Enable 0: No replacement 1: Replacement	R/W
19:17	—	These bits are read as 0. The write value should be 0.	R/W
27:20	MECS1	Map End CS1 Replace Address The bit27 to bit20 of system bus address is replaced with this value when MERCS1 is set to 1.	R/W
31:28	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

For SiP product, the MER register should be set as described in [section 44.4. SiP Product Configuration](#). For standard product, do not change the value after reset.

44.2.3 OSPI Status Registers

These registers monitor the status of xSPI Master.

44.2.3.1 COMSTT : OSPI Common Status Register

Base address: $OSPI_{n_B} = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $OSPI_{n_B_NS} = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x184

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	RSTO CS1	INTCS 1	ECSC S1	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	WRBU FNEC H0	—	PBUF NECH 0	—	—	—	MEMA CCCH 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MEMACCH0	Memory Access Ongoing from Channel 0 This bit is only valid in the section 44.3.7.2. Flow of Communication Stop and section 44.3.7.6. Flow of Memory-mapping Stop flows. 0: System bus bridge CH0 is not accessing to memory 1: System bus bridge CH0 is accessing to memory	R
3:1	—	These bits are read as 0.	R
4	PBUFNECH0	Prefetch Buffer Not Empty for Channel 0 0: Empty 1: Not empty	R
5	—	This bit is read as 0.	R
6	WRBUFNECH0	Write Buffer Not Empty for Channel 0 This bit is only valid in the section 44.3.7.2. Flow of Communication Stop and section 44.3.7.6. Flow of Memory-mapping Stop flows. 0: Empty 1: Not empty	R
19:7	—	These bits are read as 0.	R
20	ECSCS1	ECS Monitor for Slave 1 This bit indicates the value of OM_ECSINT1 port. 0: Low level 1: High level	R
21	INTCS1	INT Monitor for Slave 1 This bit indicates the value of OM_ECSINT1 port. 0: Low level 1: High level	R
22	RSTOCS1	RSTO Monitor for Slave 1 This bit indicates the value of OM_RSTO1 port. 0: Low level 1: High level	R
31:23	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

The COMSTT register indicates the status of the xSPI Master.

44.2.3.2 CASTTCSn : OSPI Calibration Status Register CSn (n = 0, 1)

Base address: $OSPI_{In_B} = 0x4026_8000 + 0x0400 \times n$ (n = 0, 1)
 $OSPI_{In_B_NS} = 0x5026_8000 + 0x0400 \times n$ (n = 0, 1)

Offset address: $0x188 + 0x004 \times n$

Bit position: 31

0

Bit field:

CASUC[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	CASUC[31:0]	Calibration Success These bits indicate the calibration success for each DS shift value. This bit is updated for each completed calibration sequence. CASUC[x] indicates calibration success in DS shift value = x.	R

Note: S-TYPE-3, P-TYPE-3

The CASTTCSn register indicates the status of xSPI Master.

44.2.4 OSPI Interrupt Registers

These registers control the interrupt function of xSPI Master.

44.2.4.1 INTS : OSPI Interrupt Status Register

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x190

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	CASU CCS1	CASU CCS0	CAFAI LCS1	CAFAI LCS0	—	—	—	—	—	—	—	BUSE RRCH 0	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	INTCS 1	—	—	—	ECSC S1	—	—	—	DSTO CS1	DSTO CS0	PERT O	—	PATC MP	CMDC MP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CMDCMP	Command Completed This bit is set to 1 when the requested manual-command is completed. In direct manual-command, it means all transactions completed. In periodic manual-command, it means the read data matches with the expected data. 0: No detection 1: Detection	R
1	PATCMP	Pattern Completed This bit is set to 1 when the requested pattern is completed. 0: No detection 1: Detection	R
2	—	This bit is read as 0.	R
3	PERTO	Periodic Transaction Timeout This bit is set to 1 when the read value does not match with the expected value in periodic manual-command mode. 0: No detection 1: Detection	R
4	DSTOCS0	DS Timeout for Slave 0 This bit is set to 1 for lost DS in read transaction when using DS. It means not receiving the data during the expected read phase. In this case, xSPI master stops the read transaction and the following transaction. This error may issue an error response to the system bus. 0: No detection 1: Detection	R
5	DSTOCS1	DS Timeout for Slave 1 This function is the same as DSTOCS0. 0: No detection 1: Detection	R
8:6	—	These bits are read as 0.	R
9	ECSCS1	ECC Error Detection for Slave 1 This bit is set to 1 when the falling edge is detected on the OM_ECSINT1 port. It can be useful only for xSPI slave with ECC detection function. 0: No detection 1: Detection	R
12:10	—	These bits are read as 0.	R
13	INTCS1	Interrupt Detection for Slave 1 This bit is set to 1 when the falling edge is detected on the OM_ECSINT1 port. It can be useful only for xSPI slave with interrupt function. 0: No detection 1: Detection	R
19:14	—	These bits are read as 0.	R

Bit	Symbol	Function	R/W
20	BUSERRCH0	System Bus Error for Channel 0 This bit is set to 1 when an error response occurs on system bus CH0.	R
27:21	—	These bits are read as 0.	R
28	CAFAILCS0	Calibration Failed for Slave 0 This bit is set to 1 when calibration failed. 0: No detection 1: Detection	R
29	CAFAILCS1	Calibration Failed for Slave 1 This function is the same as CAFAILCS0. 0: No detection 1: Detection	R
30	CASUCCS0	Calibration Success for Slave 0 This bit is set to 1 for success calibration. 0: No detection 1: Detection	R
31	CASUCCS1	Calibration Success for Slave 1 This function is the same as CASUCCS0. 0: No detection 1: Detection	R

Note: S-TYPE-3, P-TYPE-3

The INTS register indicates the status of an interrupt. The bits in this register are cleared to 0 when writing 1 on the corresponding bit of the INTC register.

44.2.4.2 INTC : OSPI Interrupt Clear Register

Base address: $\text{OSPI}_{n_B} = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $\text{OSPI}_{n_B_NS} = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x194

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	CASU CCS1 C	CASU CCS0 C	CAFAI LCS1 C	CAFAI LCS0 C	—	—	—	—	—	—	—	BUSE RRCH 0C	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	INTCS 1C	—	—	—	ECSC S1C	—	—	—	DSTO CS1C	DSTO CS0C	PERT OC	—	PATC MPC	CMDC MPC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CMDCMPC	Command Completed Interrupt Clear 0: No change interrupt status 1: Clear interrupt status	W
1	PATCMPC	Pattern Completed Interrupt Clear 0: No change interrupt status 1: Clear interrupt status	W
2	—	The write value should be 0.	W
3	PERTOC	Periodic Transaction Timeout Interrupt Clear 0: No change interrupt status 1: Clear interrupt status	W
4	DSTOCS0C	DS Timeout for Slave 0 Interrupt Clear 0: No change interrupt status 1: Clear interrupt status	W

Bit	Symbol	Function	R/W
5	DSTOCS1C	DS Timeout for Slave 1 Interrupt Clear 0: No change interrupt status 1: Clear interrupt status	W
8:6	—	The write value should be 0.	W
9	ECSCS1C	ECC Error Detection for Slave 1 Interrupt Clear 0: No change interrupt status 1: Clear interrupt status	W
12:10	—	The write value should be 0.	W
13	INTCS1C	Interrupt Detection for Slave 1 Interrupt Clear 0: No change interrupt status 1: Clear interrupt status	W
19:14	—	The write value should be 0.	W
20	BUSERRCH0C	System Bus Error for ch0 Interrupt Clear 0: No change interrupt status 1: Clear interrupt status	W
27:21	—	The write value should be 0.	W
28	CAFAILCS0C	Calibration Failed for Slave 0 Interrupt Clear 0: No change interrupt status 1: Clear interrupt status	W
29	CAFAILCS1C	Calibration Failed for Slave 1 Interrupt Clear 0: No change interrupt status 1: Clear interrupt status	W
30	CASUCCS0C	Calibration Success for Slave 0 Interrupt Clear 0: No change interrupt status 1: Clear interrupt status	W
31	CASUCCS1C	Calibration Success for Slave 1 Interrupt Clear 0: No change interrupt status 1: Clear interrupt status	W

Note: S-TYPE-3, P-TYPE-3

The INTC register clears the status of interrupt.

44.2.4.3 INTE : OSPI Interrupt Enable Register

Base address: $\text{OSPIIn_B} = 0x4026_8000 + 0x0400 \times n$ ($n = 0, 1$)
 $\text{OSPIIn_B_NS} = 0x5026_8000 + 0x0400 \times n$ ($n = 0, 1$)

Offset address: 0x198

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	CASU CCS1 E	CASU CCS0 E	CAFAI LCS1E	CAFAI LCS0E	—	—	—	—	—	—	—	BUSE RRCH 0E	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	INTCS 1E	—	—	—	ECSC S1E	—	—	—	DSTO CS1E	DSTO CS0E	PERT OE	—	PATC MPE	CMDC MPE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CMDCMPE	Command Completed Interrupt Enable 0: Disabled 1: Enabled	R/W
1	PATCMPE	Pattern Completed Interrupt Enable 0: Disabled 1: Enabled	R/W

Bit	Symbol	Function	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
3	PERTOE	Periodic Transaction Timeout Interrupt Enable 0: Disabled 1: Enabled	R/W
4	DSTOCS0E	DS Timeout for Slave 0 Interrupt Enable 0: Disabled 1: Enabled	R/W
5	DSTOCS1E	DS Timeout for Slave 1 Interrupt Enable 0: Disabled 1: Enabled	R/W
8:6	—	These bits are read as 0. The write value should be 0.	R/W
9	ECSCS1E	ECC Error Detection for Slave 1 Interrupt Enable 0: Disabled 1: Enabled	R/W
12:10	—	These bits are read as 0. The write value should be 0.	R/W
13	INTCS1E	Interrupt Detection for Slave 1 Interrupt Enable 0: Disabled 1: Enabled	R/W
19:14	—	These bits are read as 0. The write value should be 0.	R/W
20	BUSERRCH0E	System Bus Error for ch0 Interrupt Enable 0: Disabled 1: Enabled	R/W
27:21	—	These bits are read as 0. The write value should be 0.	R/W
28	CAFAILCS0E	Calibration Failed for Slave 0 Interrupt Enable 0: Disabled 1: Enabled	R/W
29	CAFAILCS1E	Calibration Failed for Slave 1 Interrupt Enable 0: Disabled 1: Enabled	R/W
30	CASUCCS0E	Calibration Success for Slave 0 Interrupt Enable 0: Disabled 1: Enabled	R/W
31	CASUCCS1E	Calibration Success for Slave 1 Interrupt Enable 0: Disabled 1: Enabled	R/W

Note: S-TYPE-3, P-TYPE-3

The INTE register enables the interrupt.

44.3 Operation

xSPI Master interface has the functions to issue the transaction for external memory with xSPI Slave interface. It allows writes to registers in external memory or reads from it.

This xSPI Master has two modes to issue transaction.

- Manual-command mode
- Memory-mapping mode

In manual-command mode, software configures all fields of xSPI frame and starts the transaction by a software request. In memory-mapping mode, software automatically converts a system bus for a preconfigured memory area into an xSPI transaction. It enables access from the system bus to the external memory area outside of the chip through xSPI bus.

This section describes the xSPI bus operation, the direct control of xSPI frame (manual-command), the control of memory access (memory-mapping), the error operation, and the flow to operation.

44.3.1 xSPI Bus

This section describes the xSPI bus operation.

Figure 44.2 shows an example of connections between OSPI and memory devices.

In this case, connect OM_CS0 to the RAM device and connect OM_CS1 to the Flash device.

Specify pull-up resistors according to the instructions for each device.

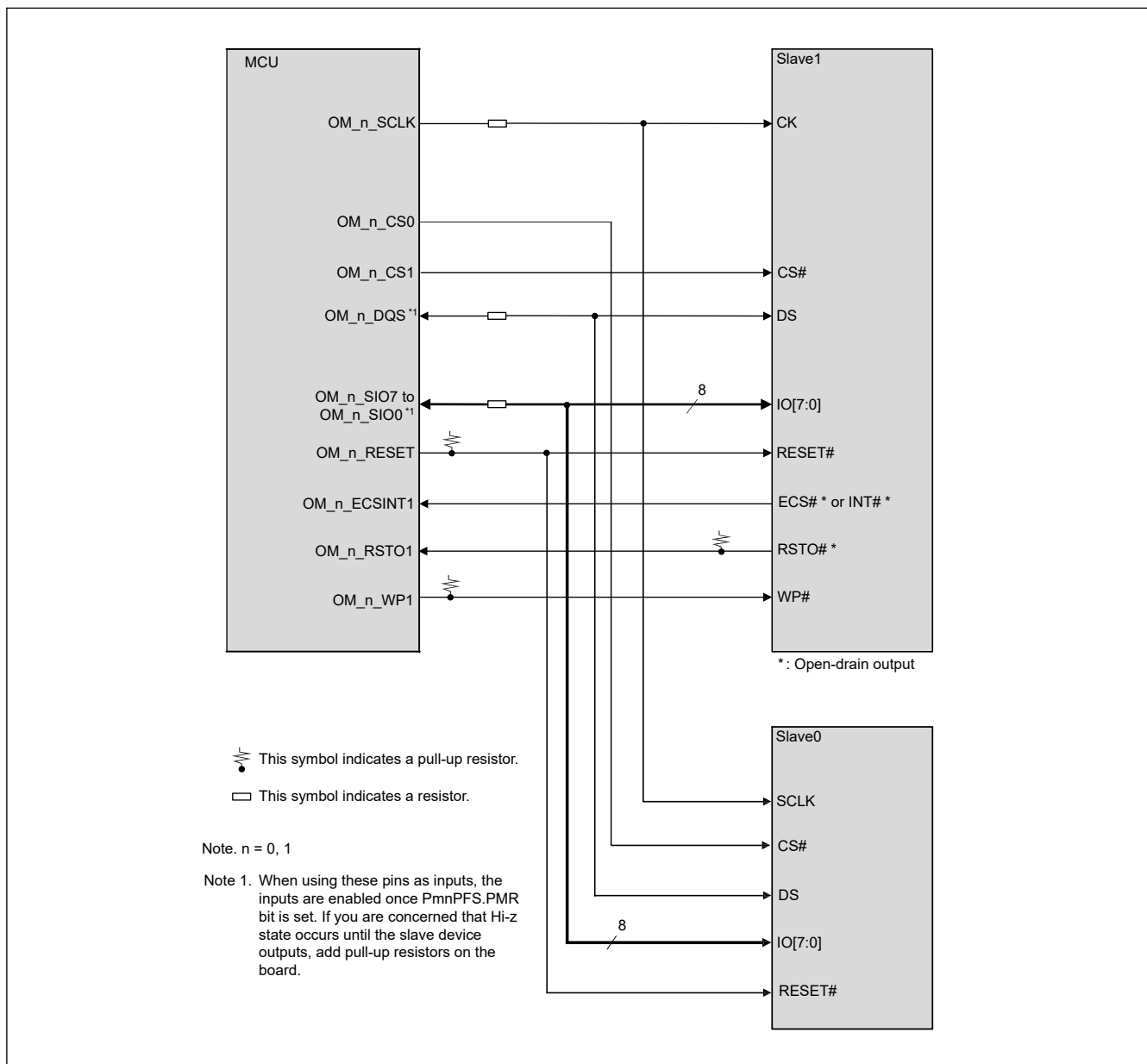


Figure 44.2 Example of connection between OSPI and memory devices

44.3.1.1 Supported Protocol Mode

This xSPI Master supports various protocol modes. It is configured by Protocol mode bits (LIOCFGCSn.PRTMD[9:0]).

Table 44.3 shows the summary of protocol modes.

Table 44.3 Supported protocol mode (1 of 2)

Protocol mode	Function	PRTMD[9:0]	Note
1S-1S-1S	Command, Address and Data fields are transferred at SDR using 1 data input pin and 1 data output pin. Read data is sampled with CK.	0x000	Specified by xSPI protocol

Table 44.3 Supported protocol mode (2 of 2)

Protocol mode	Function	PRTMD[9:0]	Note
4S-4D-4D	Command field is transferred at SDR using 4 data pins. Address and Data fields are transferred at DDR using 4 data pins. Read data is sampled with OM_DQS.	0x3B2	Specified by xSPI protocol
8D-8D-8D	Command, Address and Data fields are transferred at DDR using 8 data pins. Read data is sampled with OM_DQS.	0x3FF	Specified by xSPI protocol
1S-2S-2S	Command field is transferred at SDR using 1 data pin. Address and Data fields are transferred at SDR using 2 data pins. Read data is sampled with CK.	0x048	—
2S-2S-2S	Command, Address and Data fields are transferred at SDR using 2 data pins. Read data is sampled with CK.	0x049	—
1S-4S-4S	Command field is transferred at SDR using 1 data pin. Address and Data fields are transferred at SDR using 4 data pins. Read data is sampled with CK.	0x090	—
4S-4S-4S	Command, Address, and Data fields are transferred at SDR using 4 data pins. Read data is sampled with CK.	0x092	—

Note: In XiP mode enable, XiP code is inserted in the Latency field. It is valid only for memory-mapping mode.

The bytes of Command and Address fields are transferred in the highest order to the lowest order sequence. The sequential bytes of the Data field are transferred in the lowest address to the highest address order. When using multiple pins, the least significant bit of each byte is placed on OM_SIO0 with each higher-order bit on the successively higher numbered OM_SIO_n (n = 1 to 7) signals.

Figure 44.3 shows the timing diagram for 1S-1S-1S protocol mode. The OM_SIO0 signal is used for output data and the OM_SIO1 signal is used for input data.

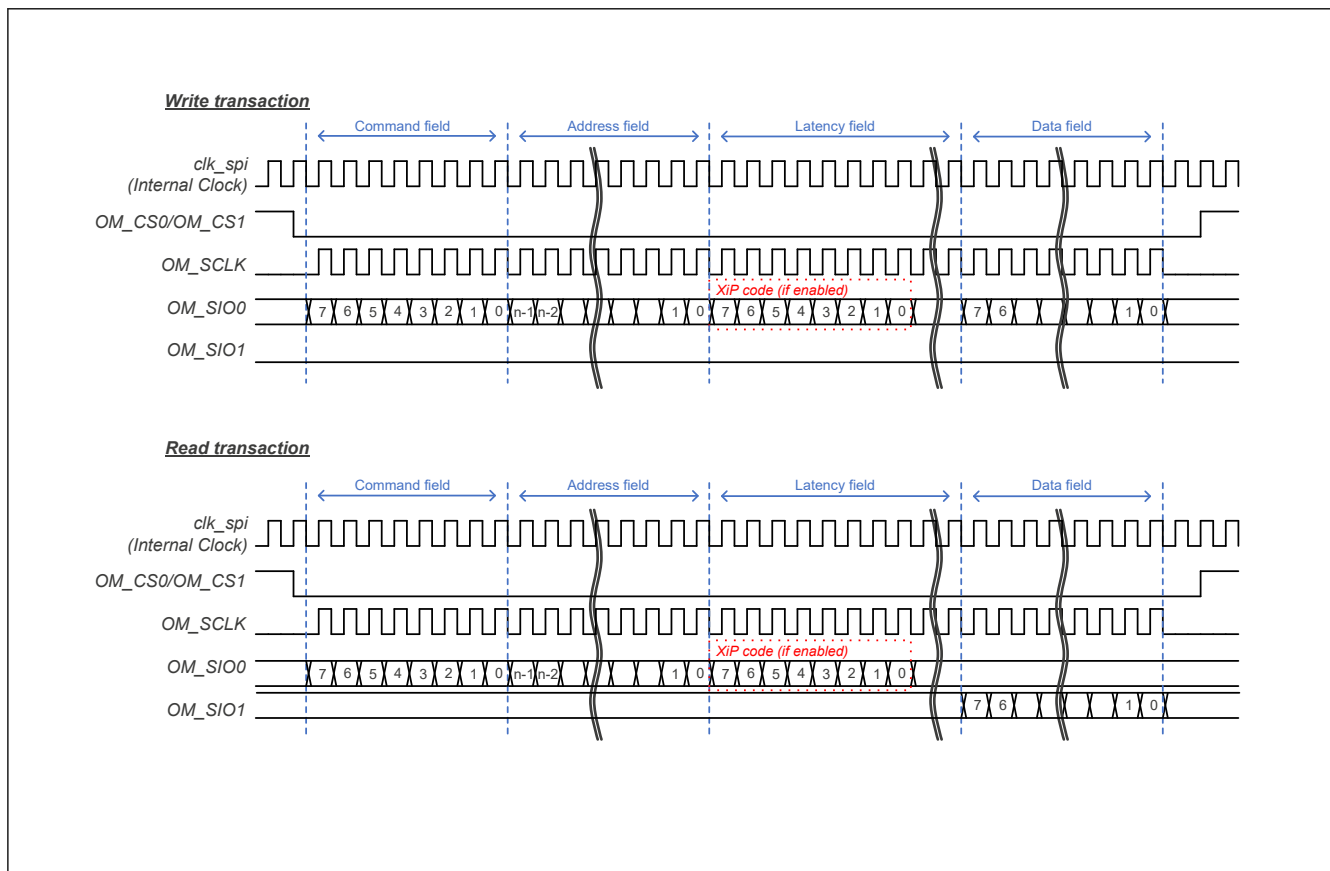


Figure 44.3 1S-1S-1S timing diagram

Figure 44.4 shows timing diagram for 1S-2S-2S protocol mode.



Figure 44.4 1S-2S-2S timing diagram

Figure 44.5 shows timing diagram for 4S-4D-4D protocol mode.

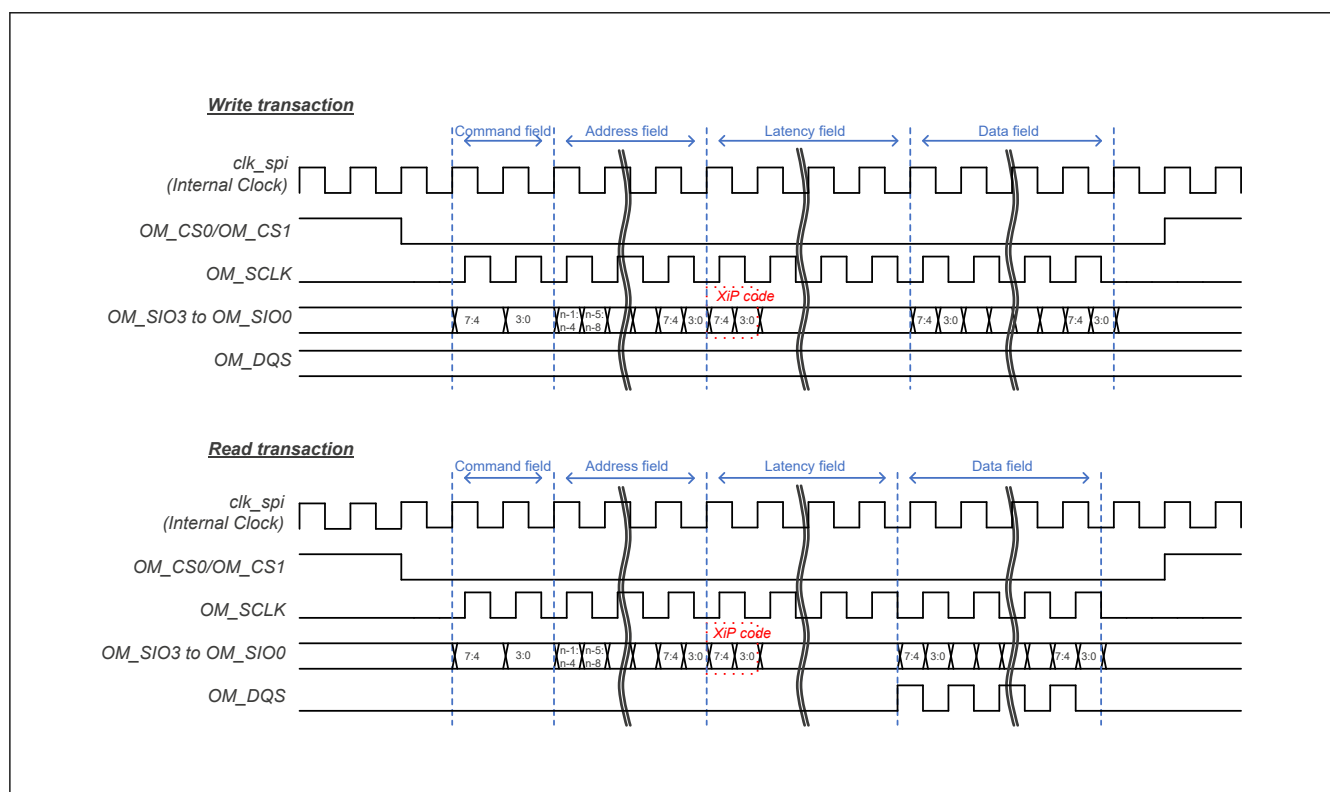


Figure 44.5 4S-4D-4D timing diagram

Figure 44.6 shows timing diagram for 8D-8D-8D profile 1.0 protocol mode. In 8D-8D-8D, the data is always transmitted with byte-pair. When each field is an odd byte, the last one byte is padded with invalid data.

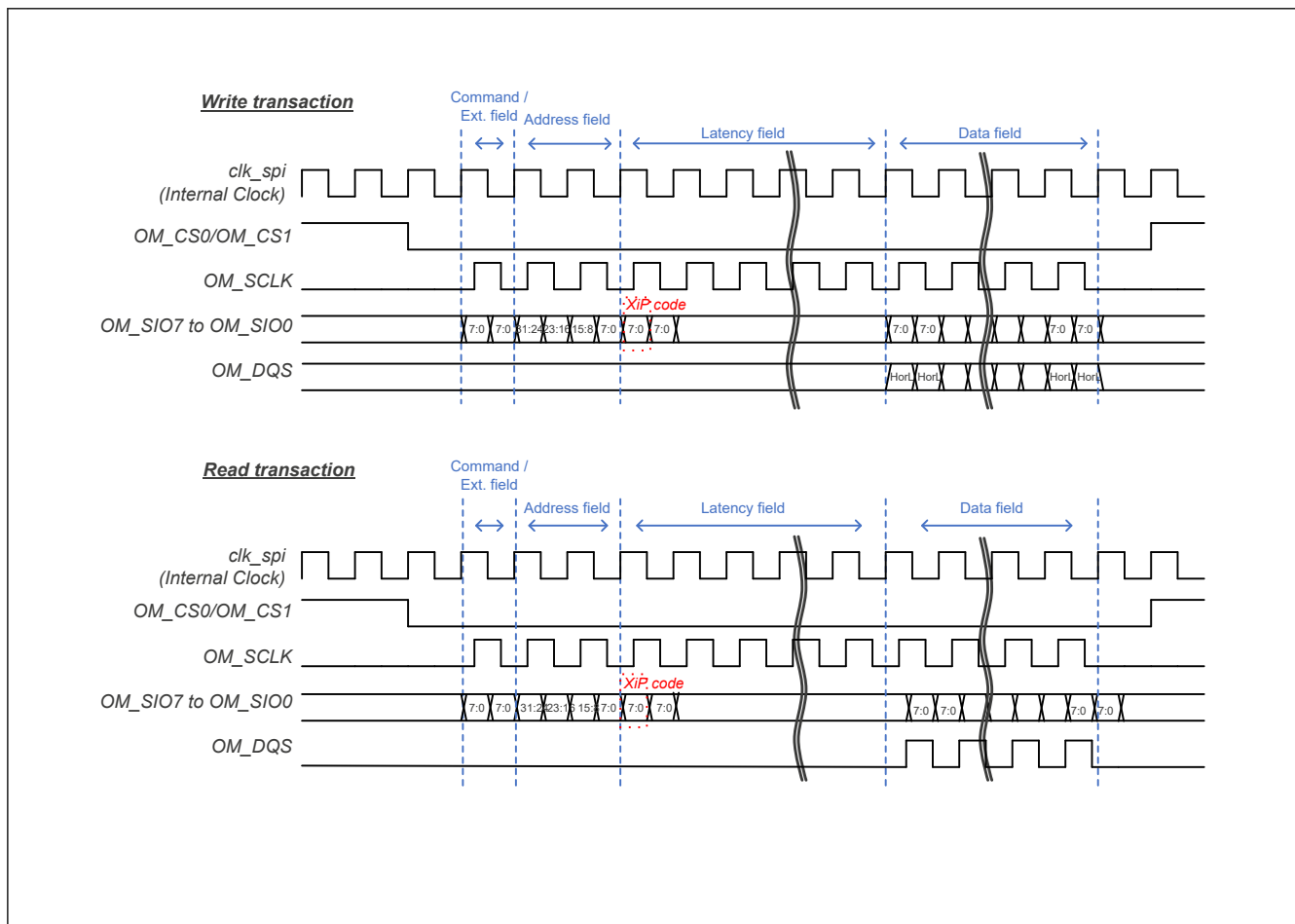


Figure 44.6 8D-8D-8D profile 1.0 timing diagram

Figure 44.7 shows timing diagram for 8D-8D-8D profile 2.0 protocol mode. In 8D-8D-8D, the data is always transmitted with byte-pair. When each field is an odd byte, the last one byte is padded with invalid data.

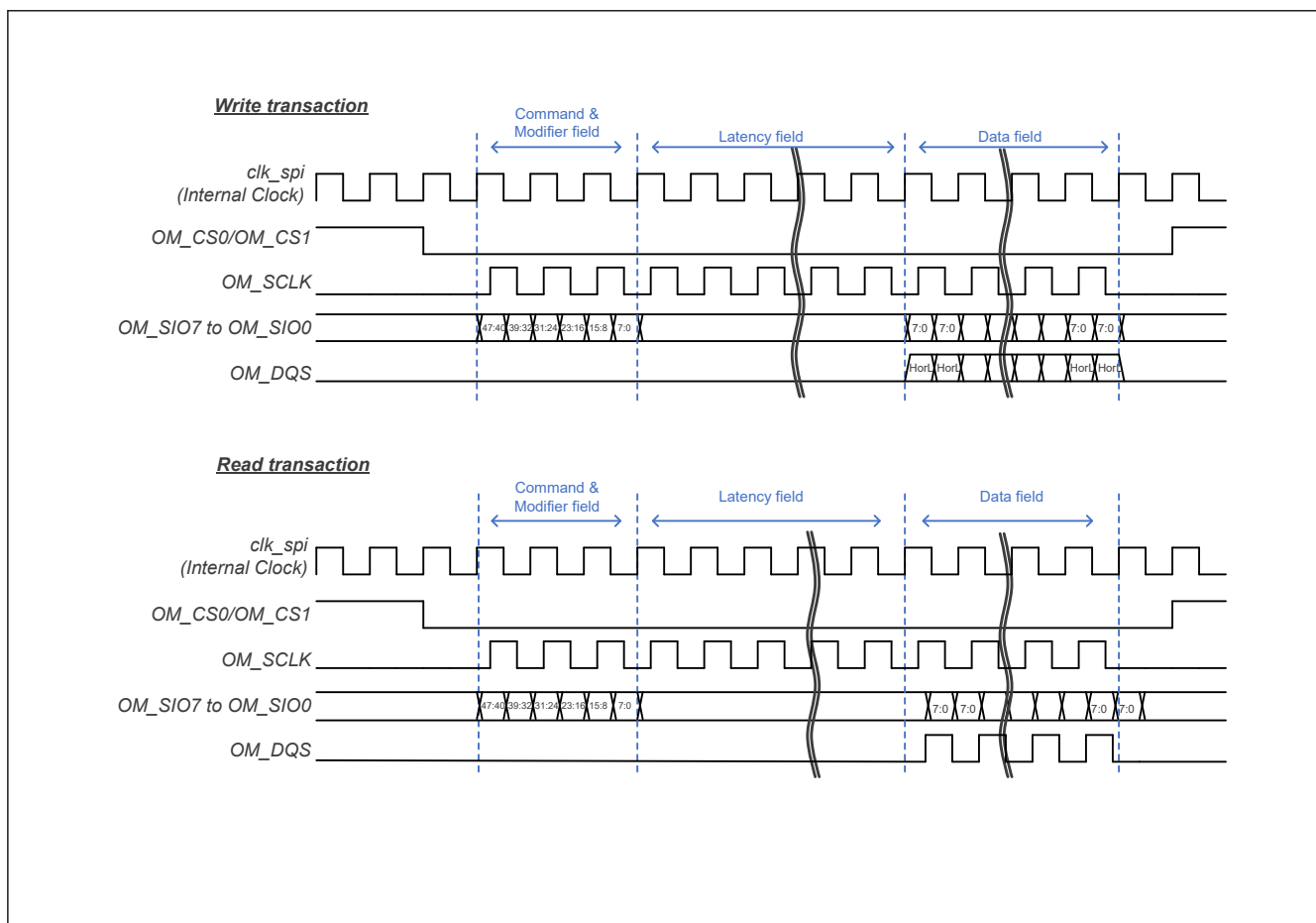


Figure 44.7 8D-8D-8D profile 2.0 timing diagram

44.3.1.2 xSPI Frame Interval

The interval between xSPI frames is configured with CS minimum idle term bits (LIOCFCGSn.CSMIN[3:0]). It depends on the specification of xSPI slave device.

44.3.1.3 OSPI Signals Timing Control

This xSPI Master supports both SDR and DDR. It is possible to sample input data with Data-Strobe (DS) signal at SDR. For various modes and easy implementation, this xSPI Master can adjust the timing to drive/sample xSPI interface signals statically. Table 44.4 shows the summary of xSPI interface signal timing control.

Table 44.4 Summary of OSPI signals timing control (1 of 2)

Signal	Mode	Default operation	Timing control (n = 0, 1)
OM_CS0, OM_CS1 drive	—	Asserting 1 cycle before the rising edge of first OM_SCLK	1 cycle extension for asserting with LIOCFCGSn.CSASTEX bit
		Negating 1.5 cycle after the falling edge of last OM_SCLK	1 cycle extension for negating with LIOCFCGSn.CSNEGEX bit
OM_SCLK drive	SDR without DS	Reference point	—
	SDR with DS		
	DDR with DS	Reference point	—

Table 44.4 Summary of OSPI signals timing control (2 of 2)

Signal	Mode	Default operation	Timing control (n = 0, 1)
OM_SIO _n (n = 0 to 7) output drive	SDR without DS	Falling edge of clk_spi (Internal Clock)	0 or 0.5 cycle shift with LIOCFGCSn.SDRDRV bit
	SDR with DS		
	DDR with DS	Both edges of clk_spi (internal clock)	—
OM_SIO _n (n = 0 to 7) input sample	SDR without DS	Falling edge of OM_SCLK on expected data size	0 to 7 cycle shift (1 cycle unit) with LIOCFGCSn.SDRSMPST[3:0] bits 0 or 0.5 cycle shift with LIOCFGCSn.SDRSMPMD bit
	SDR with DS	Falling edge of OM_DQS signal on expected data size	Sample at rising edge with LIOCFGCSn.SDRSMPMD bit 0 to 1 cycle phase shift with WRAPCFG.DSSFTCSn[4:0] bits 0 to 7 cycle extension with LIOCFGCSn.DDRSMPEX[3:0] bits
	DDR with DS	Both edges of OM_DQS signal on expected data size	0 to 1 cycle phase shift with WRAPCFG.DSSFTCSn[4:0] bits 0 to 7 cycle extension with LIOCFGCSn.DDRSMPEX[3:0] bits

Note: In DDR on xSPI protocol, DS should be aligned for the center of data. It means to shift the phase by 0.25 cycle (90 degrees). This xSPI master supports the adjustment of this phase depending on the usage conditions.

Figure 44.8 shows the default operation and timing control for SDR without DS.

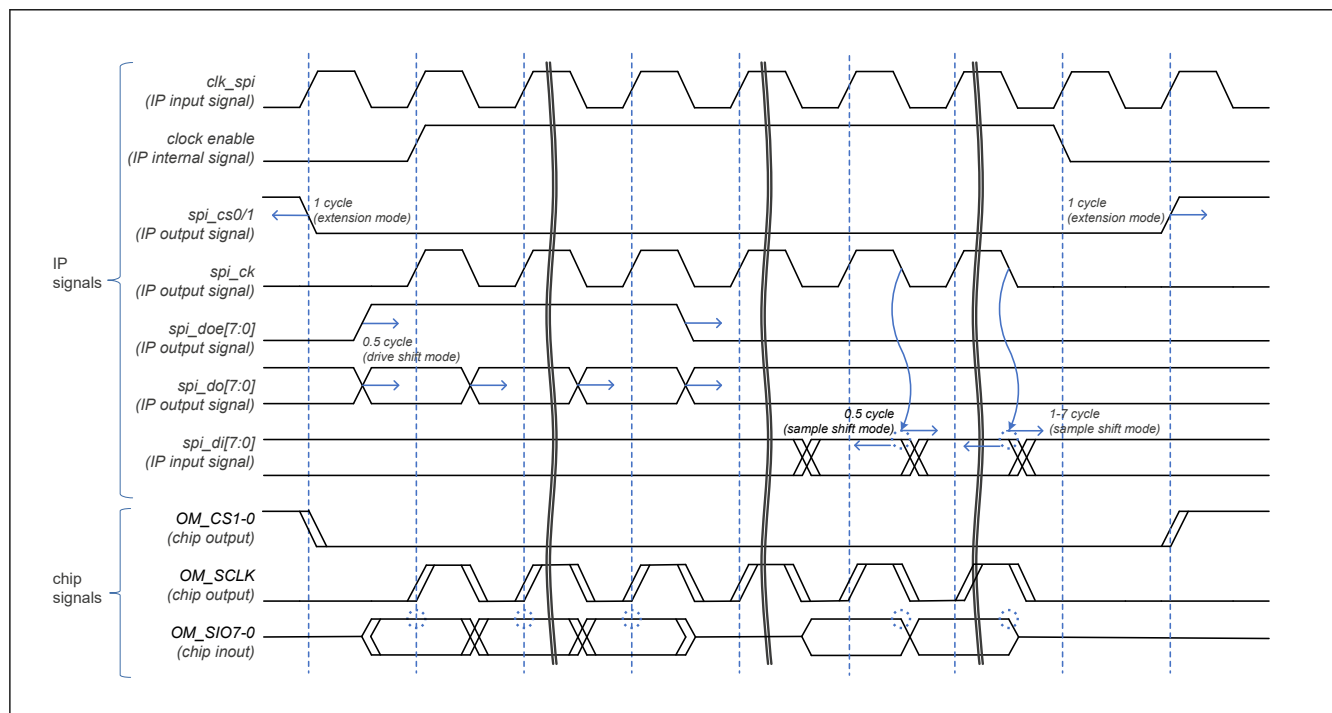


Figure 44.8 Timing control for SDR without DS

Figure 44.9 shows the default operation and timing control for SDR with DS.

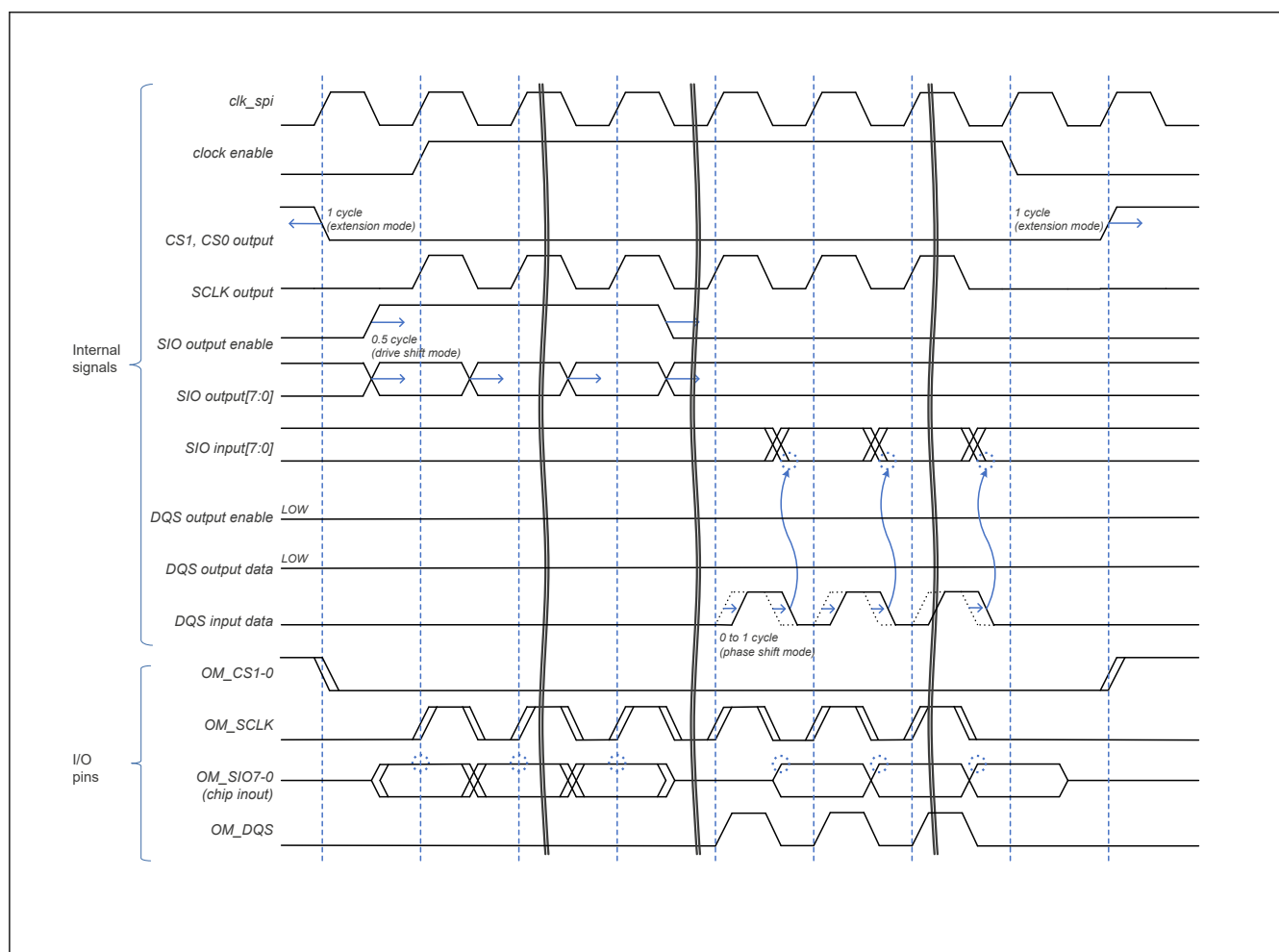


Figure 44.9 Timing control for SDR with DS

Figure 44.10 shows the default operation and timing control for DDR with DS.

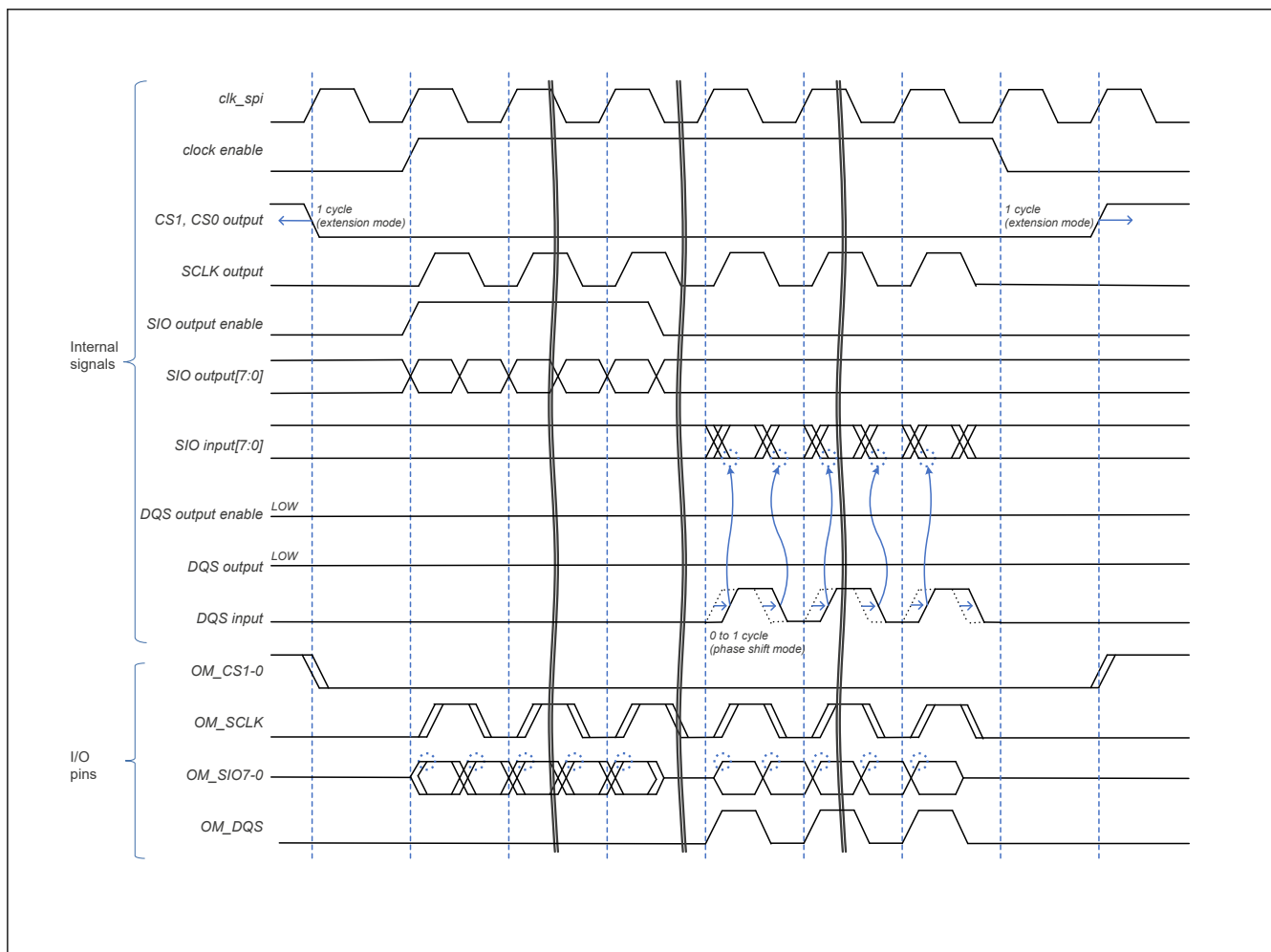


Figure 44.10 Timing control for DDR with DS

44.3.1.4 Automatic Calibration

The xSPI master supports the function to adjust DS shift value (WRAPCFG.DSSFTCSn (n = 0, 1)) automatically. When this function is enabled (CCCTL0CSn.CAEN = 1), this xSPI master transmits the calibration sequence periodically and adjusts the value of phase shift.

When all read compare is mismatched in read transaction during the calibration sequence, the calibration fail bit (INTS.CAFAILCSn = 1 (n = 0, 1)) is asserted and DS shift value is not updated. When at least one read compare is matched, the calibration success bit (INTS.CASUCCSn = 1 (n = 0, 1)) is asserted and DS shift value is updated. The result of each DS shift value can be monitored by the Calibration Status register (CASTTCSn). Figure 44.11 shows automatic calibration.

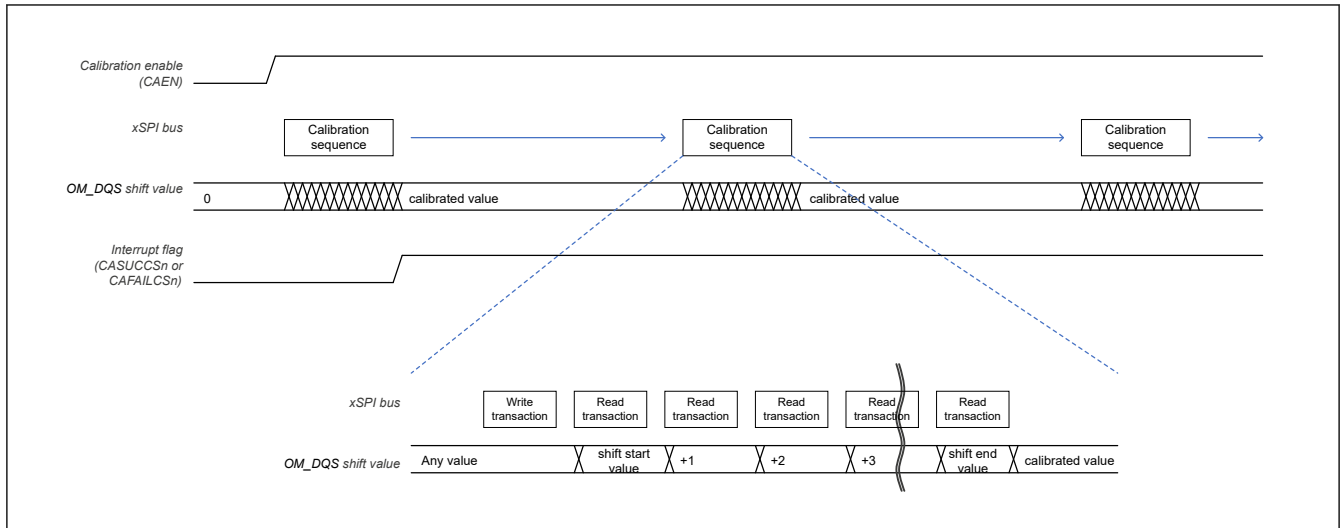


Figure 44.11 Automatic calibration

44.3.2 Manual-command

This section describes the manual-command mode. The manual-command has two functional modes:

- Direct mode
- Periodic mode.

44.3.2.1 Direct Mode

This mode sequentially can issue up to four xSPI transactions configured and requested by software. A series of transaction can be issued by a transaction request (CDCTL0.TRREQ = 1 with PERMD = 0). The number of transactions (CDCTL0.TRNUM[1:0]) can be configured up to 4. It can be used to change the mode or read the status of an xSPI slave device.

Table 44.5 shows the configured register bits for direct manual-command. Figure 44.25 shows the operating flow.

Table 44.5 Manual-command configuration for direct mode

Transaction	Transaction type	Command	Command size	Address	Address size	Data (x = 0, 1)	Data size	Latency cycle
1st Transaction	CDTBUF0. TRTYPE	CDTBUF0. CMD[15:0]	CDTBUF0. CMDSIZE [1:0]	CDABUF0. ADD[31:0]	CDTBUF0. ADDSIZE [2:0]	CDDxBUF0. DATA[31:0]	CDTBUF0. DATASIZE [3:0]	CDTBUF0. LATE[4:0]
2nd Transaction	CDTBUF1. TRTYPE	CDTBUF1. CMD[15:0]	CDTBUF1. CMDSIZE [1:0]	CDABUF1. ADD[31:0]	CDTBUF1. ADDSIZE [2:0]	CDDxBUF1. DATA[31:0]	CDTBUF1. DATASIZE [3:0]	CDTBUF1. LATE[4:0]
3rd Transaction	CDTBUF2. TRTYPE	CDTBUF2. CMD[15:0]	CDTBUF2. CMDSIZE [1:0]	CDABUF2. ADD[31:0]	CDTBUF2. ADDSIZE [2:0]	CDDxBUF2. DATA[31:0]	CDTBUF2. DATASIZE [3:0]	CDTBUF2. LATE[4:0]
4th Transaction	CDTBUF3. TRTYPE	CDTBUF3. CMD[15:0]	CDTBUF3. CMDSIZE [1:0]	CDABUF3. ADD[31:0]	CDTBUF3. ADDSIZE [2:0]	CDDxBUF3. DATA[31:0]	CDTBUF3. DATASIZE [3:0]	CDTBUF3. LATE[4:0]

44.3.2.2 Periodic Mode

This mode periodically issues an xSPI read transaction configured and requested by software, and can compare the read value up to 4 bytes with the expected value. The transaction is issued by a transaction request (CDCTL0.TRREQ = 1 with PERMD = 1). It can be used to alternate the status polling operation of an xSPI slave device.

The periodic term is configured in the periodic transaction interval bits (CDCTL0.PERITV[4:0]). The expected value is configured in the periodic transaction expected and masked value bits (CDCTL1.PEREXP[31:0] and CDCTL2.PERMSK[31:0]). Table 44.6 shows the configured register bits for periodic manual-command. Figure 44.26 shows the operating flow.

Table 44.6 Manual-command configuration for periodic mode

Transaction	Transaction type	Command	Command size	Address	Address size	Data (x = 0, 1)	Data size	Latency cycle
Read Transaction	CDTBUF0. TRTYPE	CDTBUF0. CMD[15:0]	CDTBUF0. CMSIZE [1:0]	CDABUF0. ADD[31:0]	CDTBUF0. ADDSIZE [2:0]	CDDxBUF0. DATA[31:0]	CDTBUF0. DATASIZE [3:0]	CDTBUF0. LATE[4:0]

44.3.3 Memory-mapping

This section describes the memory-mapping mode. This mode automatically converts system bus access for a preconfigured memory area into an xSPI transaction.

44.3.3.1 Configuration

In this operation, the payload of the Address and Data fields are delivered from system bus signals. The information of the Command field and size are delivered from the configured register bits. [Table 44.7](#) shows the register bits configured for memory-mapping.

Table 44.7 Memory-mapping configuration for memory area access (n = 0, 1)

System bus transaction	Format change mode	Command	Command size	Address	Address size	Data	Data size	Latency cycle
Write for slave n memory area	Normal	CMCFG2CSn. WRCMD[15:8]	1 byte	SAWADDR[x:0]	CMCFG0C Sn. ADDSIZE [1:0]	SWDATA	Up to SAWLEN and SAWSIZE	CMCFG2CSn. WRLATE[4:0]
	8D-8D-8D profile 1.0	CMCFG2CSn. WRCMD[15:0]	2 bytes					
	8D-8D-8D profile 2.0 Command Modifier	CMCFG2CSn. WRCMD[15:8]	1 byte	{SAWADDR[27:4], 0000000000000b, SAWADDR[3:1]}	5 bytes			
	8D-8D-8D profile 2.0 Extended Command Modifier	CMCFG2CSn. WRCMD[15:13]	3 bits	{0b, SAWADDR [31:4], 0000000000000b, SAWADDR[3:1]}	45 bits			
Read for slave n memory area	Normal	CMCFG1CSn. RDCMD[15:8]	1 byte	SARADDR[x:0]	CMCFG0C Sn. ADDSIZE [1:0]	SRDATA	Up to SARLEN and SARSIZE	CMCFG1CSn. RDLATE[4:0]
	8D-8D-8D profile 1.0	CMCFG1CSn. RDCMD[15:0]	2 bytes					
	8D-8D-8D profile 2.0 Command Modifier	CMCFG1CSn. RDCMD[15:8]	1 byte	{SARADDR[27:4], 0000000000000b, SARADDR[3:1]}	5 bytes			
	8D-8D-8D profile 2.0 Extended Command Modifier	CMCFG1CSn. RDCMD[15:13]	3 bits	{0b, SARADDR [31:4], 0000000000000b, SARADDR[3:1]}	45 bits			

Note: The MSByte of Address can be replaced with Address Replace Enable and Code bits (CMCFG0CSn.ADDRPEN[7:0]/ADDRPCD[7:0]).

44.3.3.2 Write Access Operation

At accepting write access for memory area from system bus, this xSPI Master stores all payload data in an internal bridge buffer and then issues a write transaction into an xSPI slave. [Figure 44.12](#) shows the operation summary.

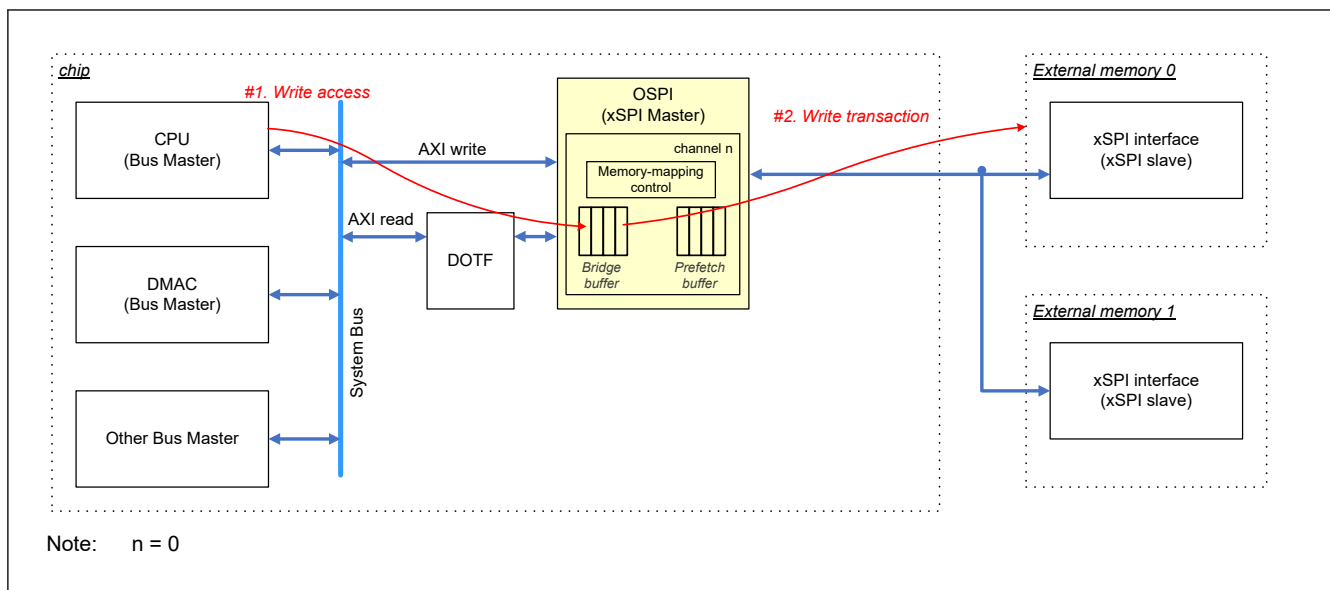


Figure 44.12 Write access for memory area

The operation of xSPI bus changes depending on the burst type of the system bus. When the burst type is single type or incremental type, one system bus transaction triggers one xSPI frame. When the burst type is wrap type and the CMCFG0CSn.WPBSTMD is 0, one system bus transaction triggers two xSPI frames. [Figure 44.13](#) shows the relationship between AXI and xSPI frames.

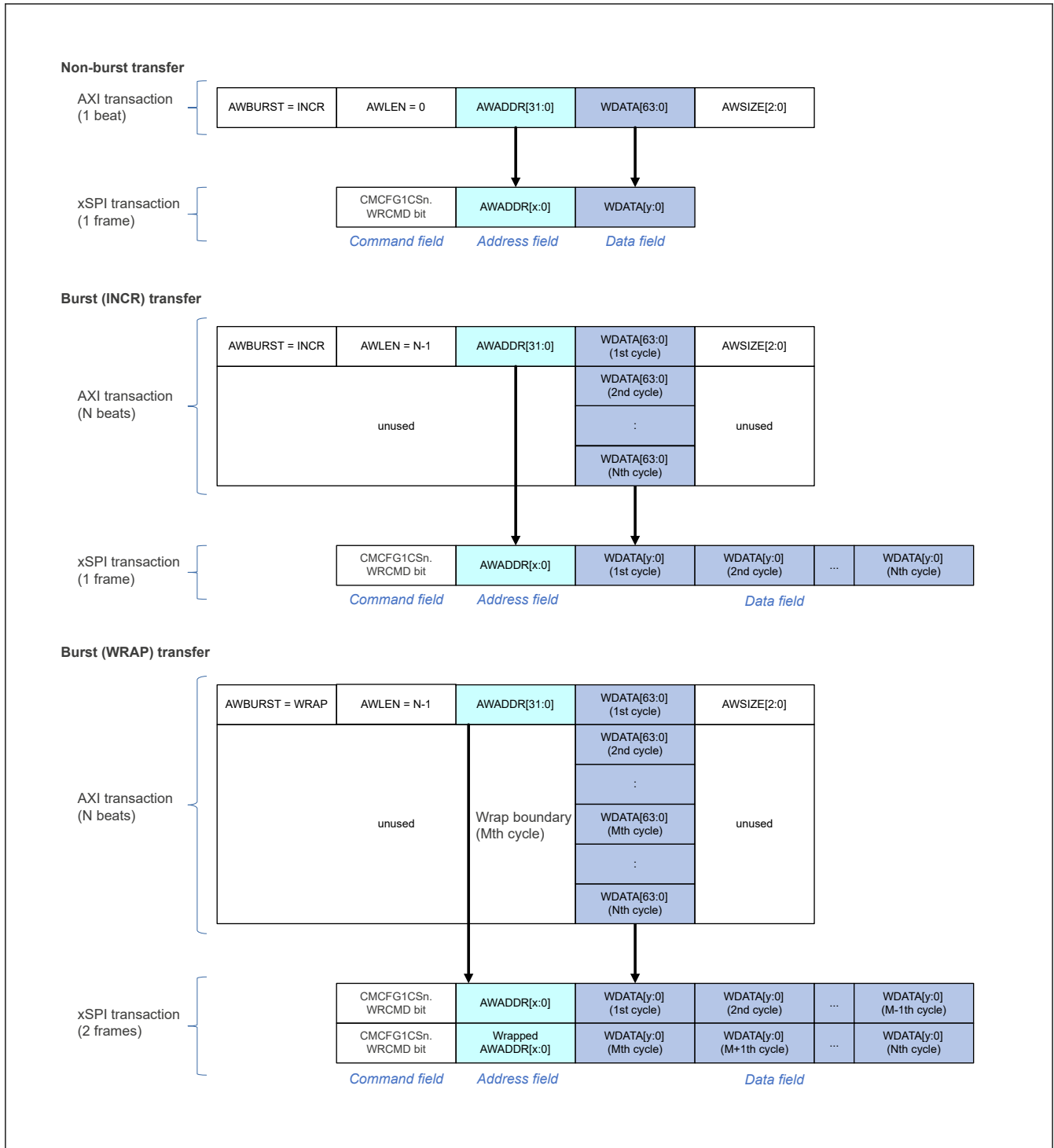


Figure 44.13 xSPI frame format in write access (Normal format)

44.3.3.3 Combination Function

At the system bus write access for memory area, this xSPI master has the function to combine the write data for high throughput on the xSPI bus. When this function is enabled (BMCFGCHn.MWRCOMB = 1), this xSPI master transmits a xSPI frame with the selected size while the sequential address is incremental^{*1}. When one of the below conditions is detected, even though not reaching to the target size (BMCFGCHn.MWRSIZE[7:0]), this xSPI master transmits the pending data into the xSPI bus.

- Nonincremental address is detected
- Different burst type is detected

- Read transaction is detected
- Access for different slave is detected
- Memory Write Data Push bit (BMCTL1.MWRPUSHCHn (n = 0)) is set.

This function is useful for any slave device to request a chunk of data at a time. In such case, the system bus master continues to provide the fixed data size with an incremental address. For example, there is a device to request to write in page unit.

Figure 44.14 shows the operation when enabling the combination function.

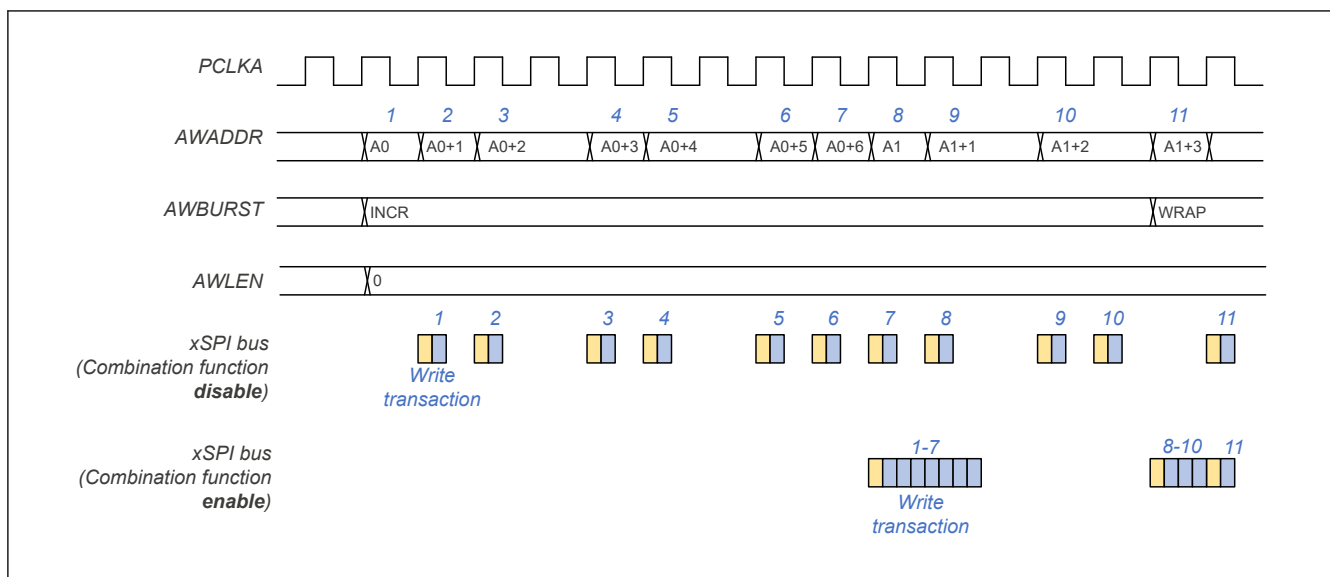


Figure 44.14 Combination function

Note 1. The access which complies with all the following conditions is considered as incremental address.

- Transaction type is INCR
- The start address of the access is continuous to the previous last write address.
 - The start position of the access for WSTRB is treated as the start address
 - The last WSTRB of the previous access is treated as the last write address.

Figure 44.15 shows a data combined example with AXI access.

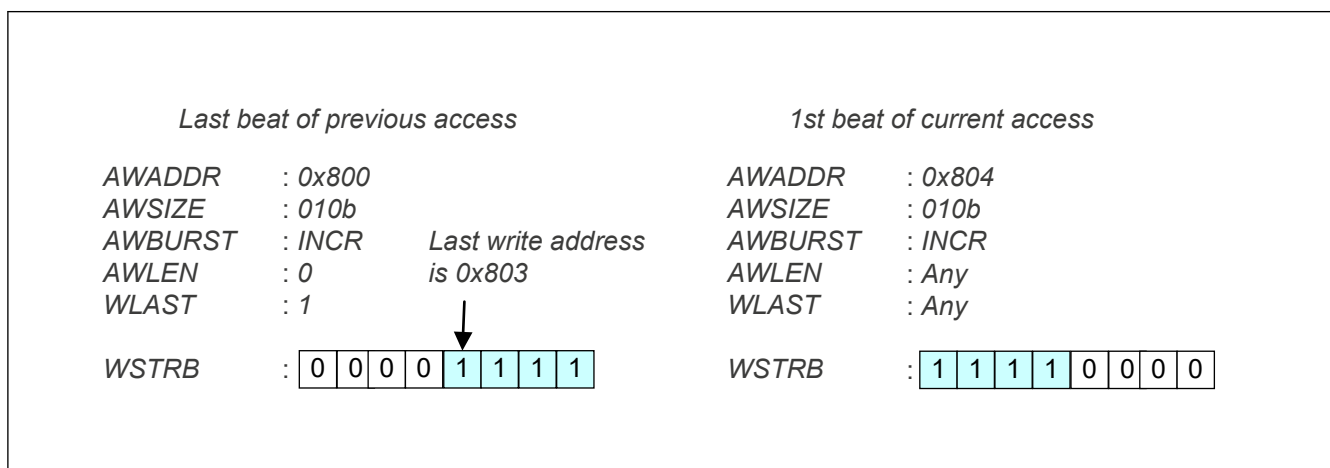


Figure 44.15 Data combined example with AXI access

44.3.3.4 Read Access Operation

At the read access for memory area, soon after detected the read access, this xSPI Master issues a read transaction into an xSPI slave. Figure 44.16 shows the operation summary.

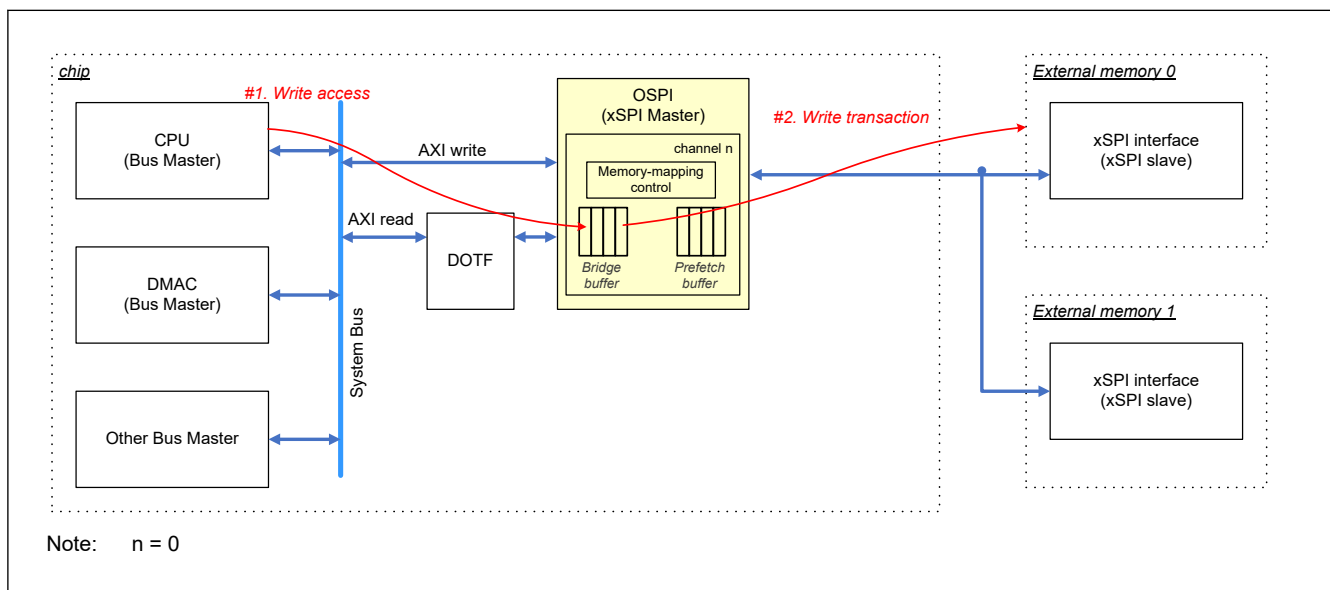


Figure 44.16 Read access for memory area

The operation of xSPI bus changes depending on the burst type. When the type is single or increment type, one read transaction of a system bus triggers one xSPI frame. When the type is wrap type and the CMCFG0CSn.WPBSTMD is 0, one read transaction of a system bus triggers two xSPI frames. Figure 44.17 shows the relationship between AXI and xSPI frames.

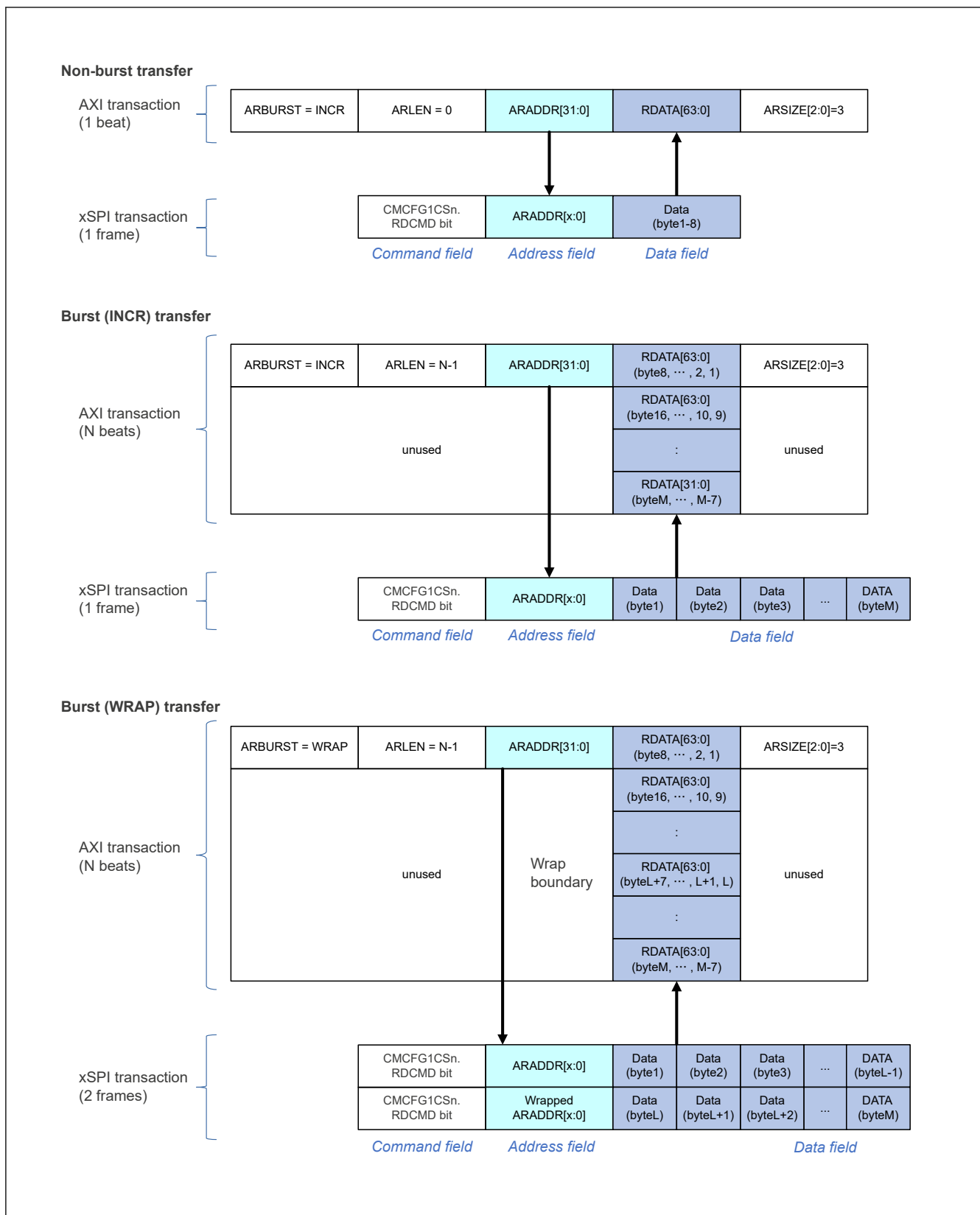


Figure 44.17 xSPI frame format in read access (Normal format)

44.3.3.5 Prefetch Function

At the read access for memory area from the system bus, this xSPI Master has the function to prefetch read data for reducing the latency. When enabled this function (BMCFGCHn.PREEN = 1), this xSPI Master continues to read the

incremental address and store the read data from an xSPI slave in internal prefetch buffer. This xSPI master then searches in the prefetch buffer for the following read access from the system bus. If the target read data in the prefetch buffer is found, this xSPI Master returns the data from the prefetch buffer. If not found, this xSPI Master clears the prefetch buffer and issues a new read transaction into an xSPI slave. This function is effective in applications such as when the consecutive read addresses are close. But it is not effective in applications such as when the consecutive read addresses are not incremental because xSPI read frame for prefetch uses xSPI bus. Figure 44.18 shows the operation summary.

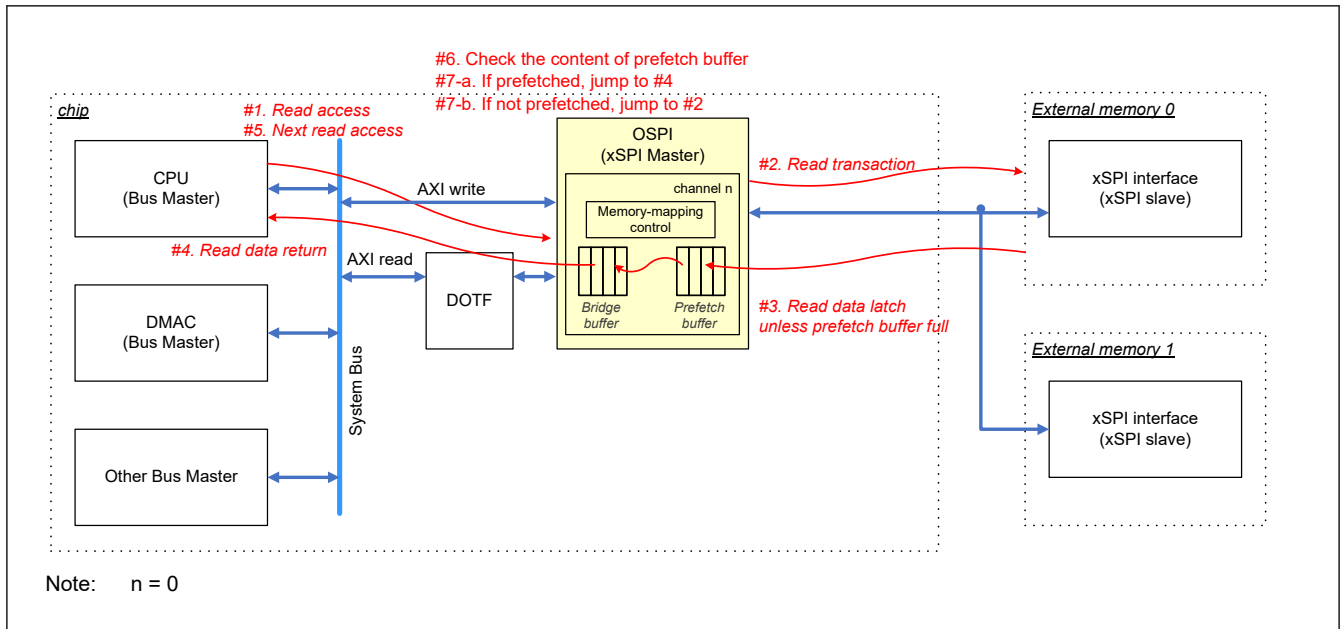


Figure 44.18 Read access for memory area with prefetch enabled

Note: When enabled this prefetch function, the bus master can read from the internal prefetch buffer but not from a slave device. When accessed to the same address from multiple bus masters, this xSPI Master does not guarantee to read the latest data. If the bus master wants to read the latest data from a slave device, it should read after clearing the prefetch buffer (BMCTL1.PBUFCLRCHn ($n = 0$)).

Note: Prefetch buffer is implemented as FIFO-based, and when a read access is issued, the data before the access address is discarded from the buffer. When the next access is issued to the region which is discarded at the previous access, this module issues the xSPI read access again to fill the prefetch buffer.

Note: This IP has a 1-line buffer which keeps the last 8-byte read data from the prefetch buffer. When a read access is issued to the same address and to the data in the 1-line buffer, this IP returns read data from the 1-line buffer.

44.3.3.6 XiP Mode

Some slave devices have a mode (XiP mode) in which the command phase is not required for lower latency. While in this mode, the xSPI master skips sending the command and the slave device implicitly performs the command that was executed in the previous transaction. When enabled XiP mode bit (CMCTLCHn.XIPEN = 1), this xSPI master inserts XiP enter code (CMCTLCHn.XIPEXCODE[7:0]) in the Latency field. When disabled XiP mode bit (CMCTLCHn.XIPEN = 0), this xSPI master inserts XiP exit code (CMCTLCHn.XIPEXCODE[7:0]) in the Latency field. This function is available only for memory-mapping mode.

When the xSPI master transmits the XiP disable pattern, this master clears the XiP mode bit and disables XiP mode configured for both channels. It is not possible to disable XiP mode for only one channel by transmitting a XiP disable pattern.

Note: When enough latency cycle does not exist for XiP code, this xSPI Master cannot insert XiP code.

Note: XiP mode can be used only for unidirectional access to a slave. The write transaction and read transaction should be separated.

Note: The XiP exit code is inserted once when disabled. For more details, see Figure 44.30.

44.3.4 Pattern Control

This xSPI Master has the function to transmit three types of pattern which are not xSPI frame format. The pattern is triggered by the setting trigger bit (LPCTL0-1.PATREQ).

44.3.4.1 XiP Disable Pattern

XiP Disable pattern transmits any pattern with the configured length and value (LPCTL0.XD1LEN[4:0]/XD1VAL/XD2LEN[4:0]/XD2VAL). XiP Disable pattern uses OM_SCLK, OM_SIO7-0 signals. The number of output pins can be configured by the XiP Disable pattern pin bits (LPCTL0.XDPIN[1:0]). It may be used to disable XiP mode for legacy SPI. [Figure 44.19](#) shows the timing diagram.

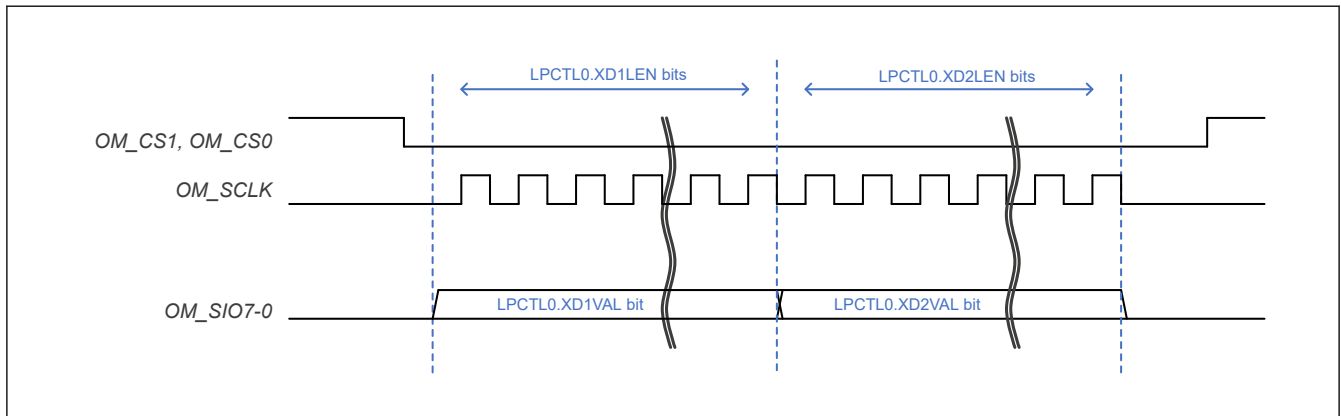


Figure 44.19 XiP Disable pattern

44.3.4.2 Reset Pattern

Reset pattern transmits the pattern specified in the Serial Flash Reset Signaling Protocol. [Figure 44.20](#) shows the timing diagram.

CS Low/High width is configured with the Reset Pattern Length bits (LPCTL1.RSTWID[2:0]). The xSPI slave samples the data input at the rising edge of CS. Setup time for data output is configured with Reset pattern data output setup time bits (LPCTL1.RSTSU[2:0]). The setup time should always be less than Reset pattern width.

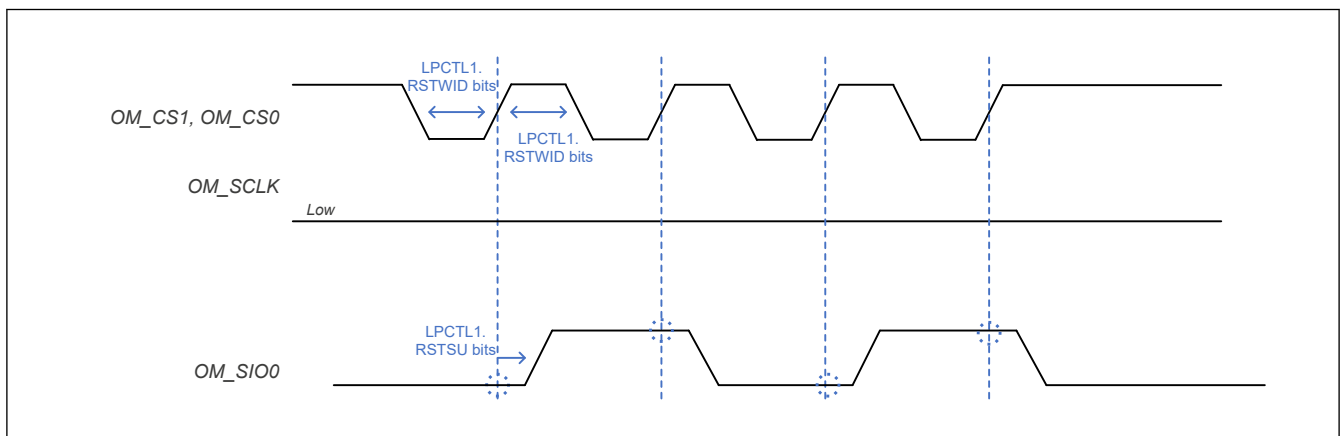
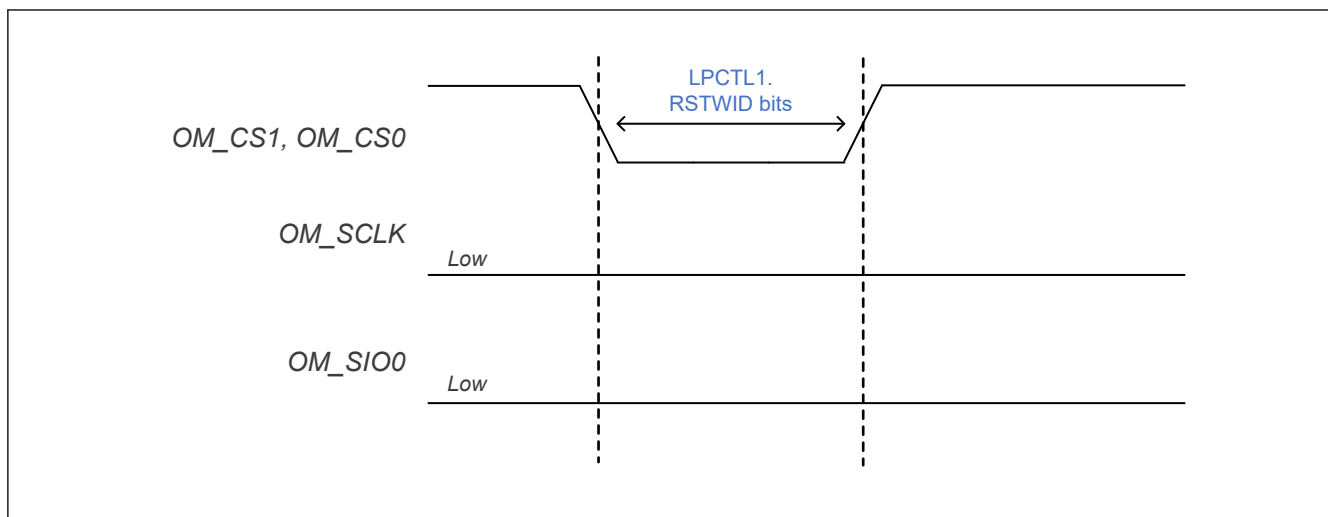


Figure 44.20 Reset pattern

Note: In the protocol, CS Low/High width is defined as minimum 500 ns and setup time is defined as minimum 6 ns.

44.3.4.3 CS-only Pattern

CS-only pattern activates the CS port with the configured length bits (LPCTL1.RSTWID[2:0]). It may be used to resume from the Deep Power Down state. [Figure 44.21](#) shows the timing diagram.

**Figure 44.21 CS-only pattern**

44.3.5 Integrity Checking

This xSPI Master can detect some errors. [Table 44.8](#) shows the error list and the detail behavior.

Table 44.8 Error list (n = 0, 1)

Error type	Event	Flag bit	Note (action)
Calibration failed	When the read data did not match the expected value during automatic calibration.	INTS.CAFAILCSn	It results in writing unexpected data to xSPI slave.
System bus error	When an error response occurred on the AXI slave interface for memory-mapping.	INTS.BUSERRCHn	This xSPI master should be reset for fatal error.
ECC error detection	When detected the falling edge on OM_ECSINT1 port. It can be useful only for xSPI slave with ECC detection function.	INTS.ECSCS1	Only notify the error event of xSPI slave.
DS timeout	When DS does not toggle in read transaction with using DS.	INTS.DSTOCSn	If this error occurs during calibration, clear the interrupt status bit. Other than above, both xSPI master and xSPI slave should be reset for fatal error.
Periodic transaction timeout	When the read value does not match with the expected value in periodic manual-command mode.	INTS.PERTO	Depending on the status of the function.

44.3.6 Interrupts

This xSPI Master has an interrupt port. It can monitor with the Interrupt Status Register (INTS). During the initialization phase, it can be programmable with the Interrupt Enable register (INTE). [Table 44.9](#) shows OSPI interrupt sources, and [Table 44.10](#) shows the related register bit.

Note: The interrupt pulse port signal is not asserted when the corresponding bit of INTE is set after the interrupt event is detected.

Table 44.9 OSPI interrupt sources

Name	Interrupt sources	DMAC activation
OSPI0_ERR	Error	Not possible
OSPI0_CMP	Complete	Not possible

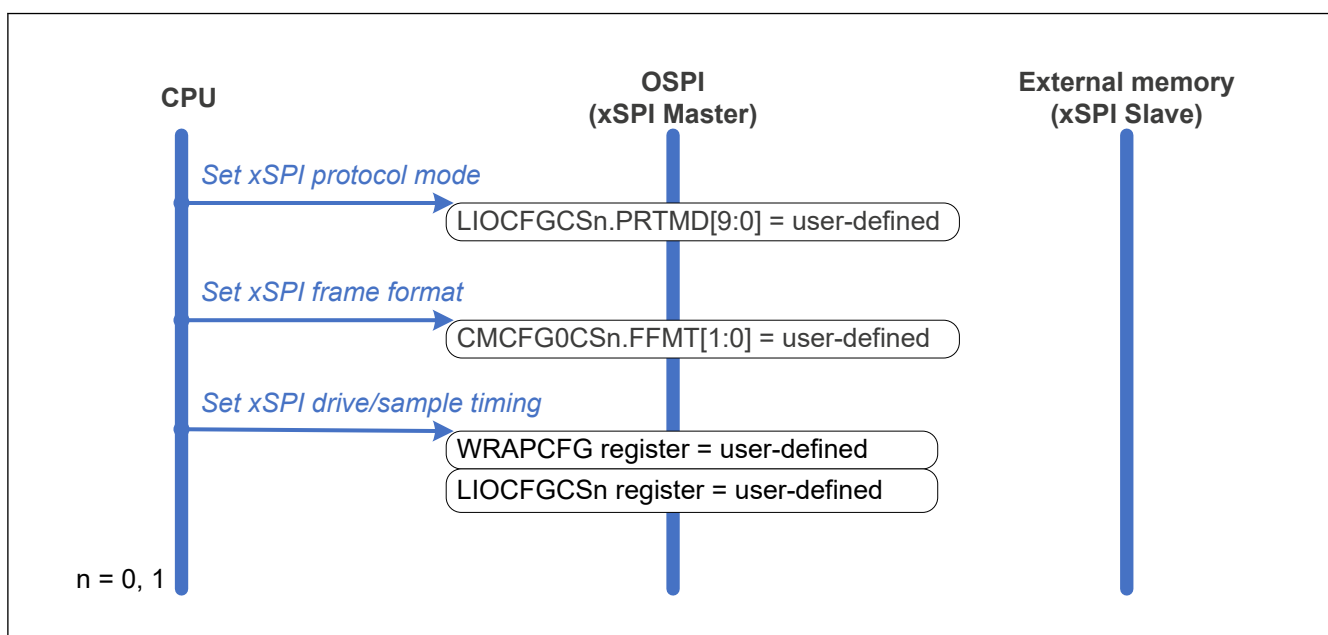
Table 44.10 Interrupt register bit

Flag bit	Enable bit	Clear bit	Interrupt sources
CASUCCS1	CASUCCS1E	CASUCCS1C	OSPI0_CMP
CASUCCS0	CASUCCS0E	CASUCCS0C	
CAFAILCS1	CAFAILCS1E	CAFAILCS1C	OSPI0_ERR
CAFAILCS0	CAFAILCS0E	CAFAILCS0C	
BUSERRCH0	BUSERRCH0E	BUSERRCH0C	OSPI0_ERR
INTCS1	INTCS1E	INTCS1C	OSPI0_ERR
ECSCS1	ECSCS1E	ECSCS1C	OSPI0_ERR
DSTOCS1	DSTOCS1E	DSTOCS1C	OSPI0_ERR
DSTOCS0	DSTOCS0E	DSTOCS0C	
PERTO	PERTOE	PERTOC	OSPI0_ERR
PATCMP	PATCMPE	PATCMPC	OSPI0_CMP
CMDCMP	CMDCMPE	CMDCMPC	OSPI0_CMP

44.3.7 Flows of Operations

44.3.7.1 Flow of Configuration

Figure 44.22 shows flow of configuration.

**Figure 44.22** Flow of configuration

44.3.7.2 Flow of Communication Stop

Figure 44.23 shows the flow of a communication stop.

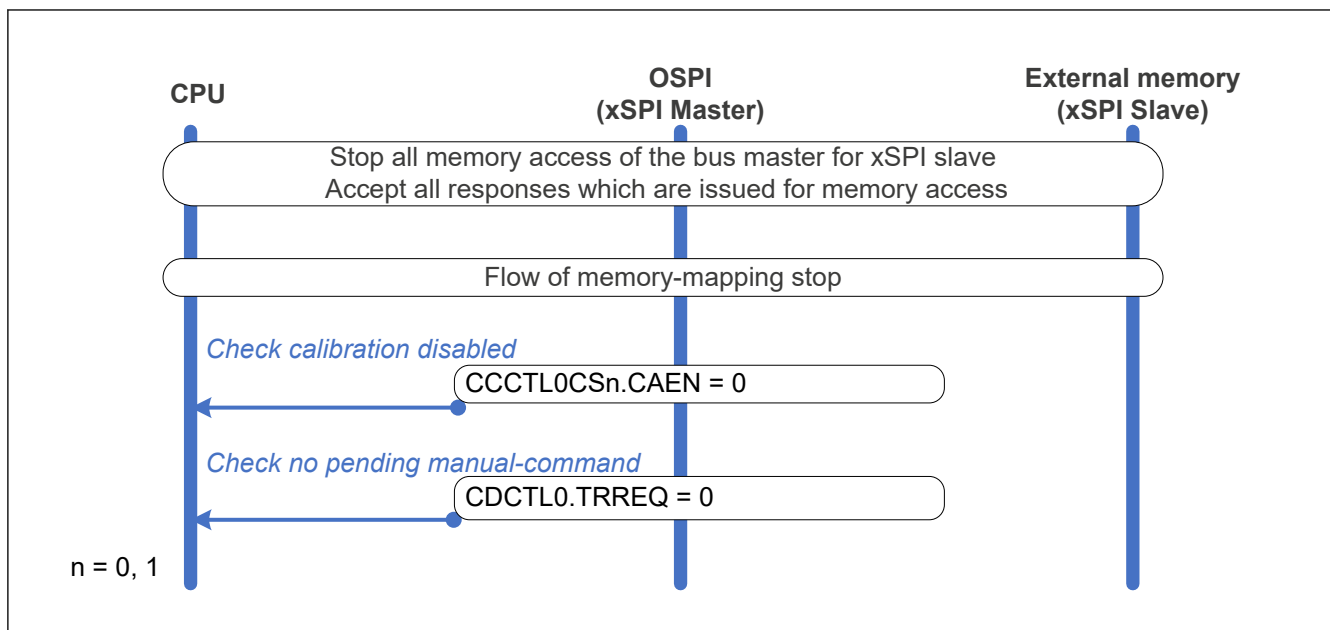


Figure 44.23 Flow of communication stop

Note: When reconfiguring any configuration register, stop all communication with the xSPI slave to avoid race condition between register setting and memory access. It means the automatic calibration is disabled, and there is no pending manual-command and memory-mapping access.

44.3.7.3 Flow of Automatic Calibration

Figure 44.24 shows the flow of an automatic calibration.

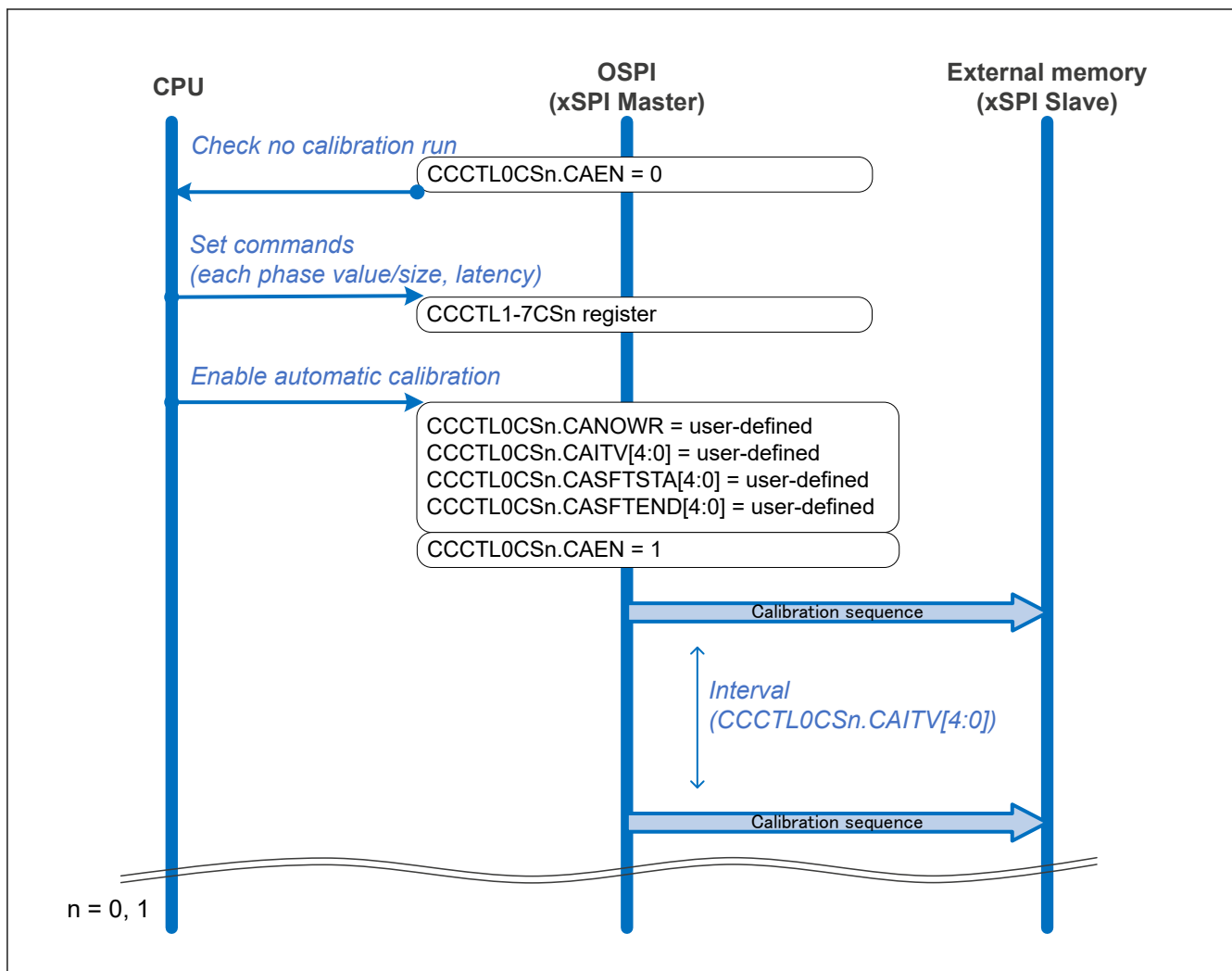


Figure 44.24 Flow of automatic calibration

44.3.7.4 Flow of Manual-command Procedure

Figure 44.25 shows the manual-command procedure for direct mode.

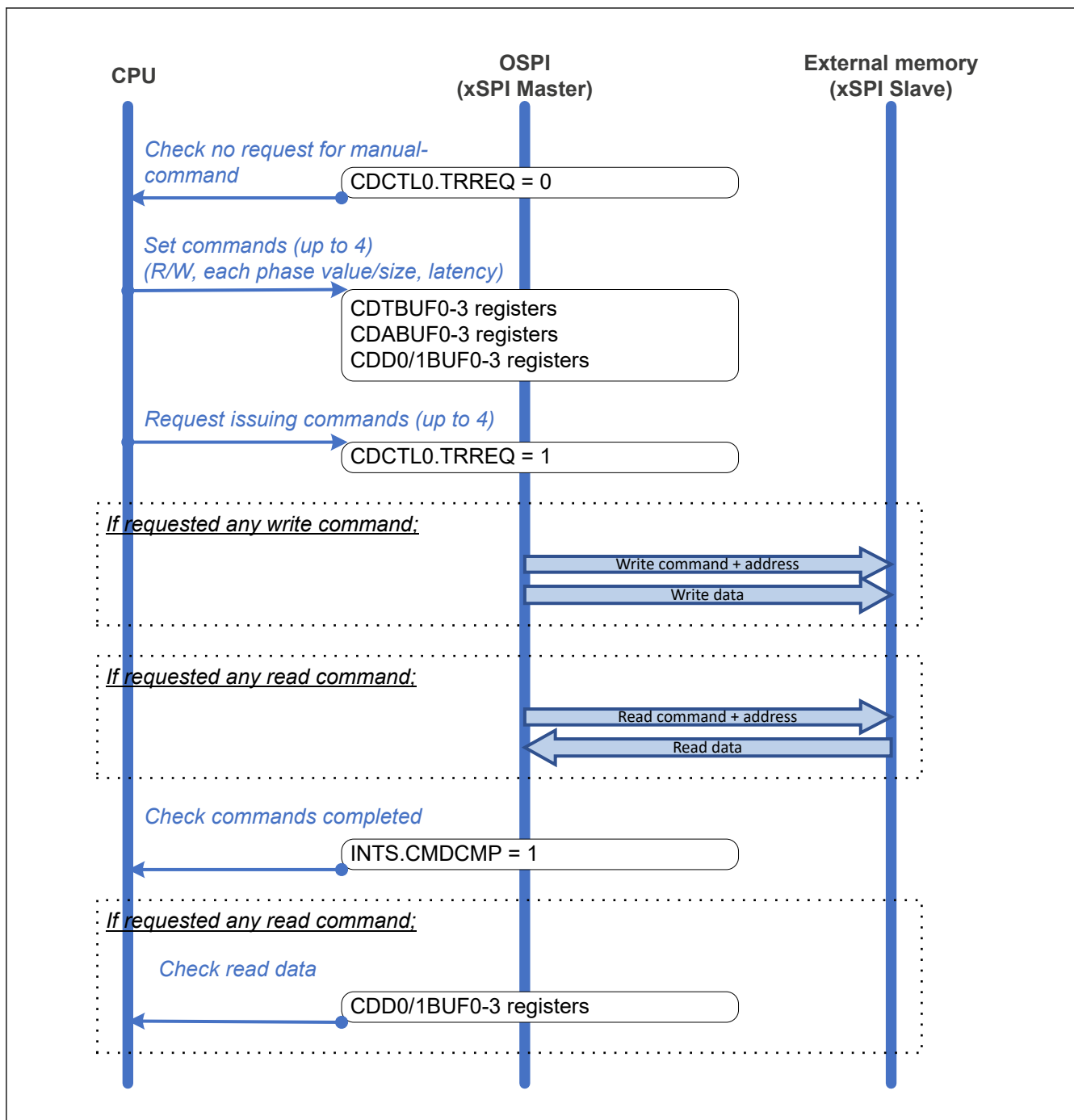


Figure 44.25 Flow of manual-command procedure for direct mode

Figure 44.26 shows the manual-command procedure for periodic mode.

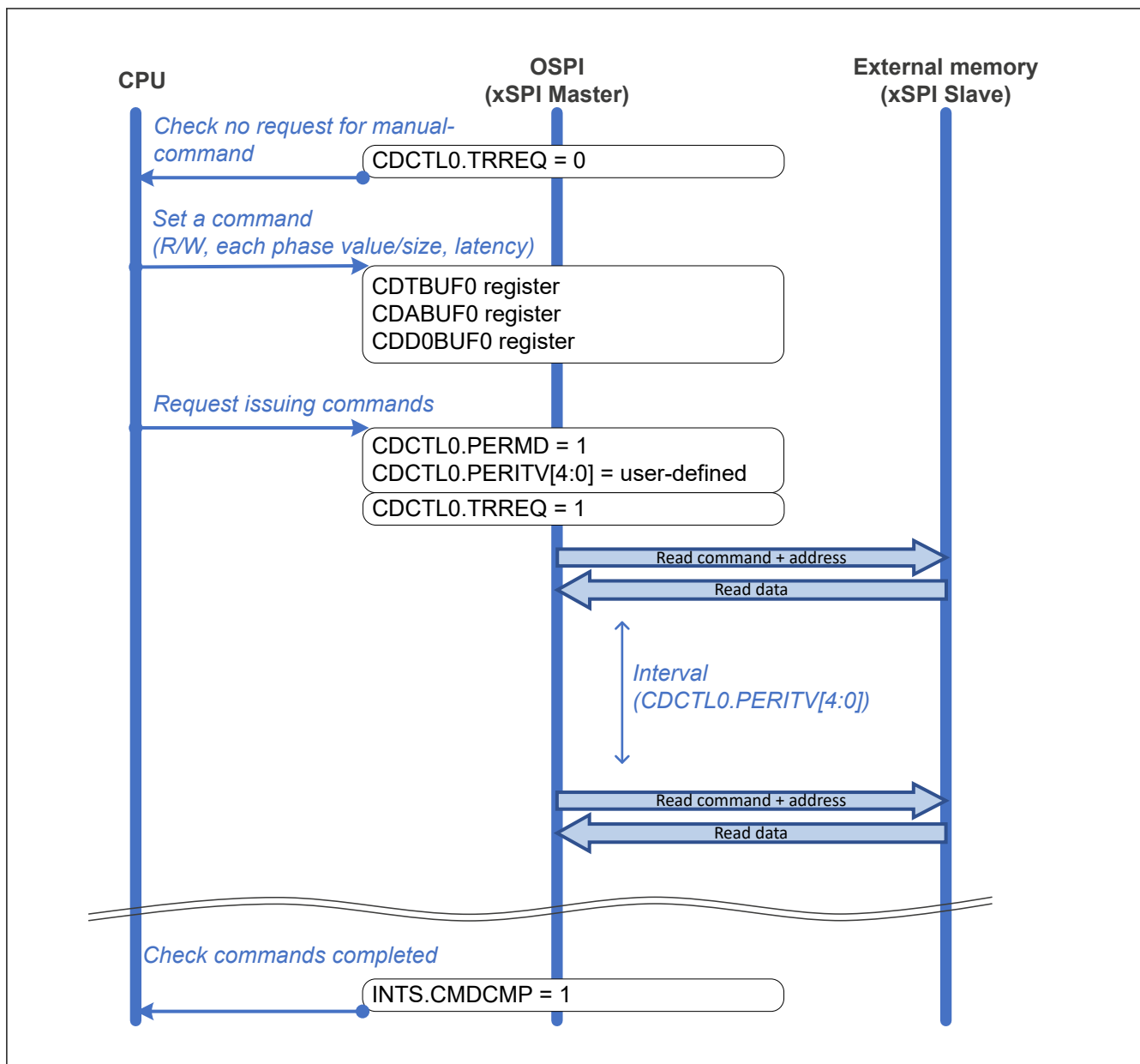


Figure 44.26 Flow of manual-command procedure for periodic mode

44.3.7.5 Flow of Memory-mapping

Figure 44.27 shows the flow of a memory-mapping.

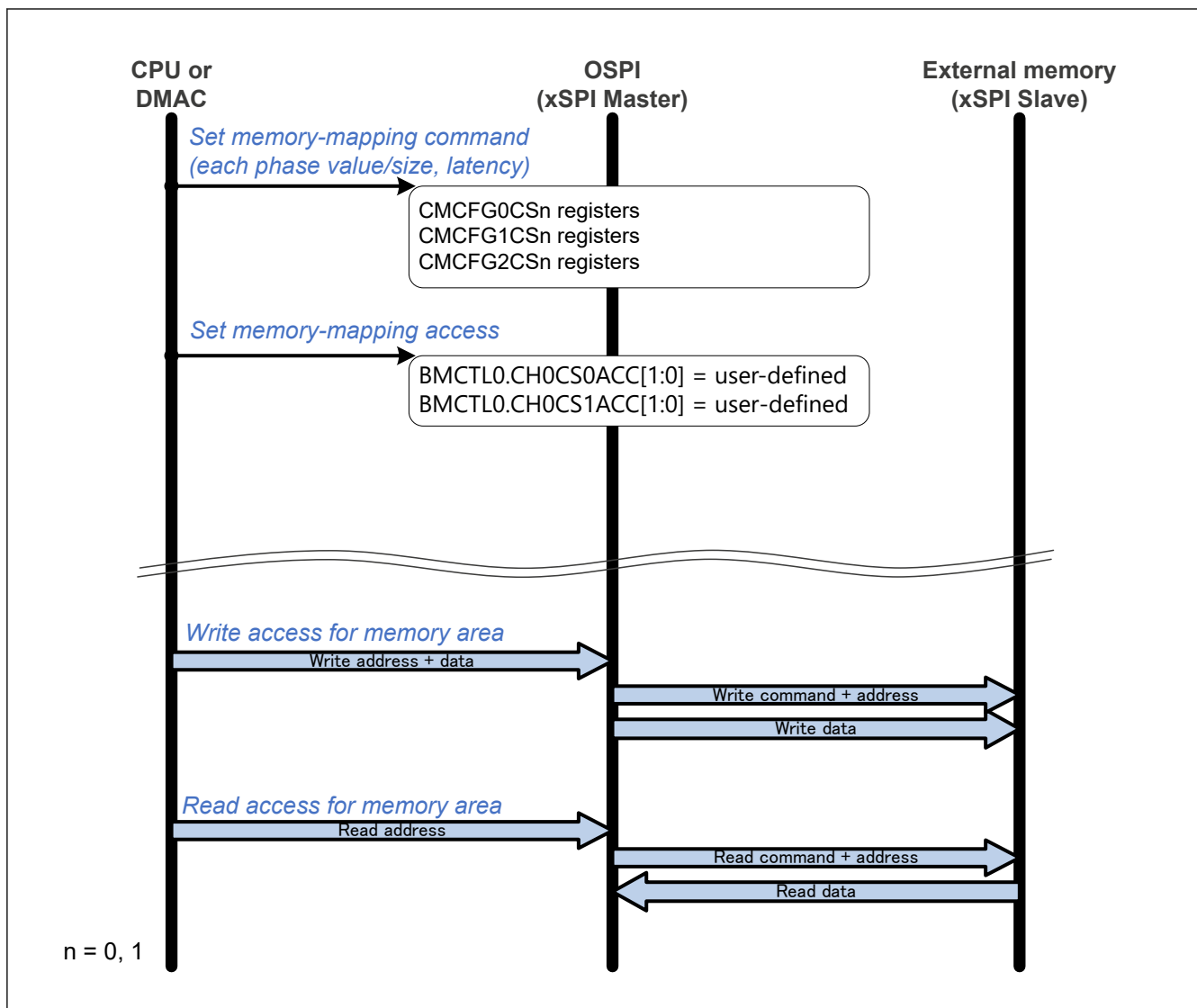


Figure 44.27 Flow of memory-mapping

44.3.7.6 Flow of Memory-mapping Stop

Figure 44.28 shows the flow of a memory-mapping stop.

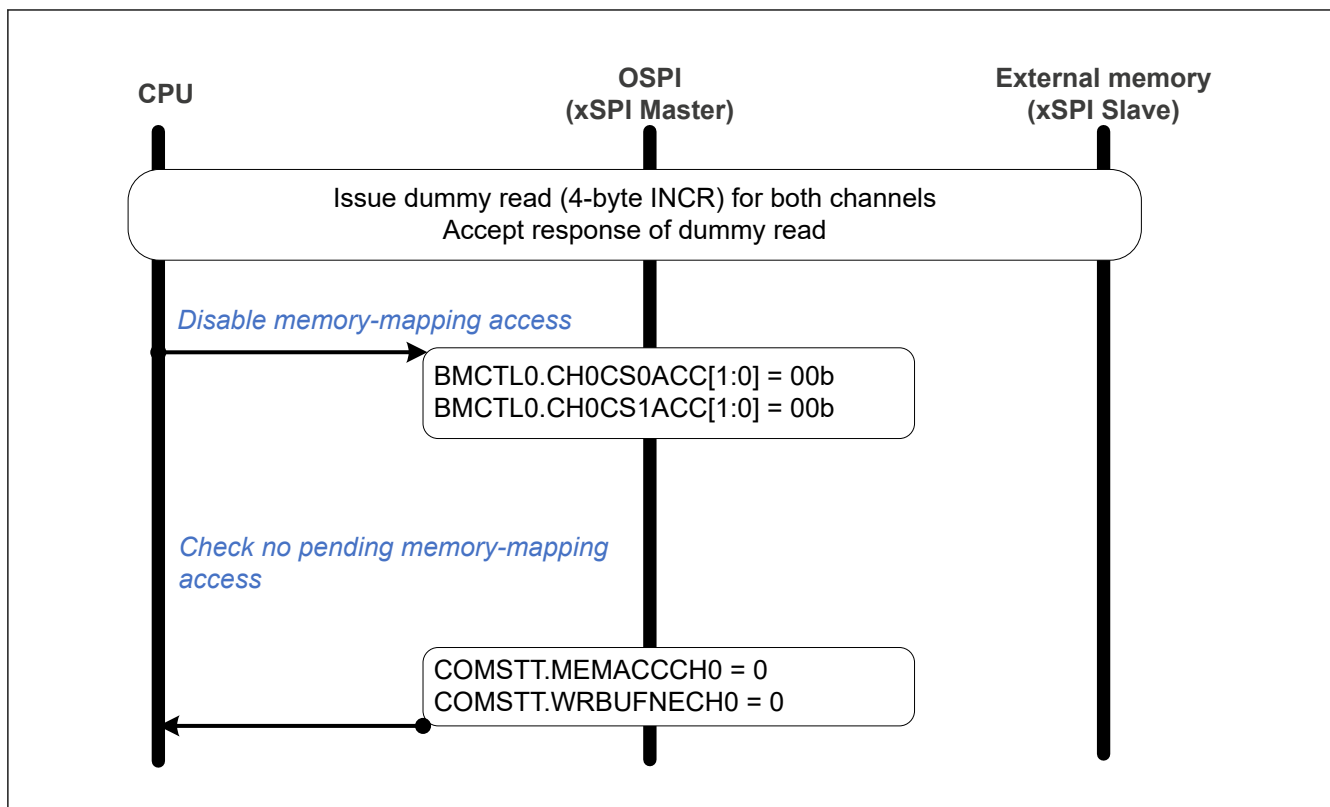


Figure 44.28 Flow of memory-mapping stop

44.3.7.7 Flow of Pattern Request

Figure 44.29 shows the flow of a pattern request. Before requesting any pattern, any ongoing commands should be completed or canceled.

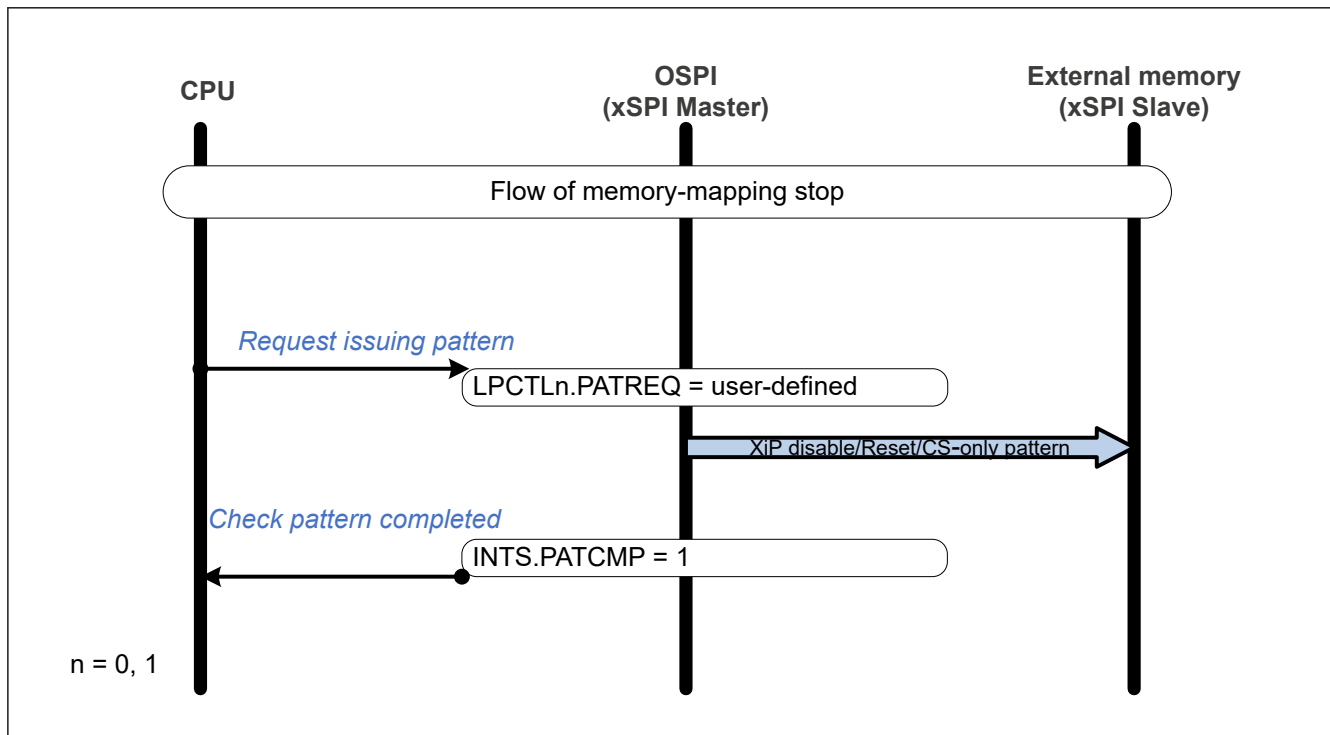


Figure 44.29 Flow of pattern request

44.3.7.8 Flow of XiP Mode

Figure 44.30 shows the flow of XiP mode enable/disable.

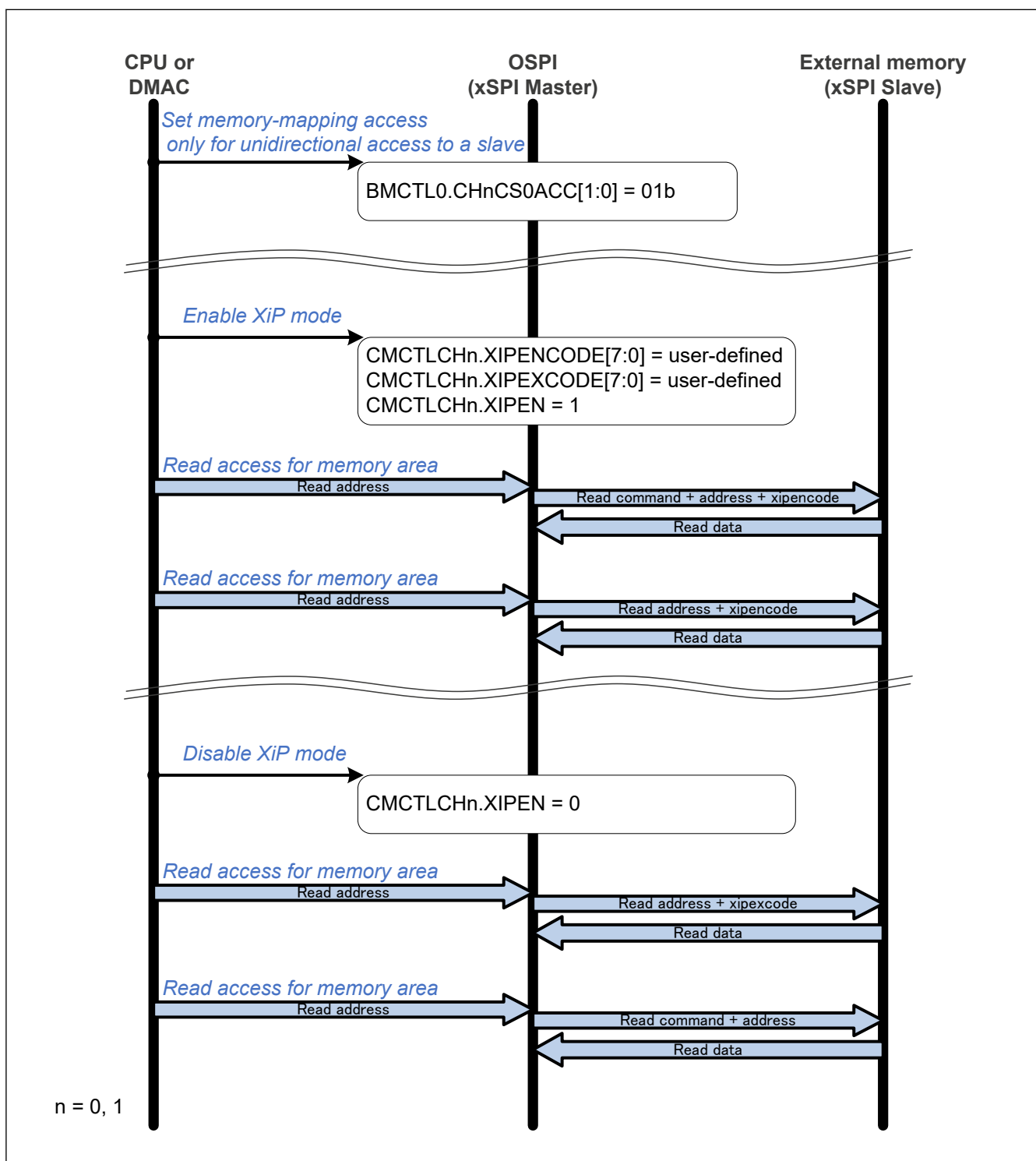


Figure 44.30 Flow of XiP mode enable/disable

44.3.7.9 Flow of Read While Write

Figure 44.31 shows the example flow of a read while write.

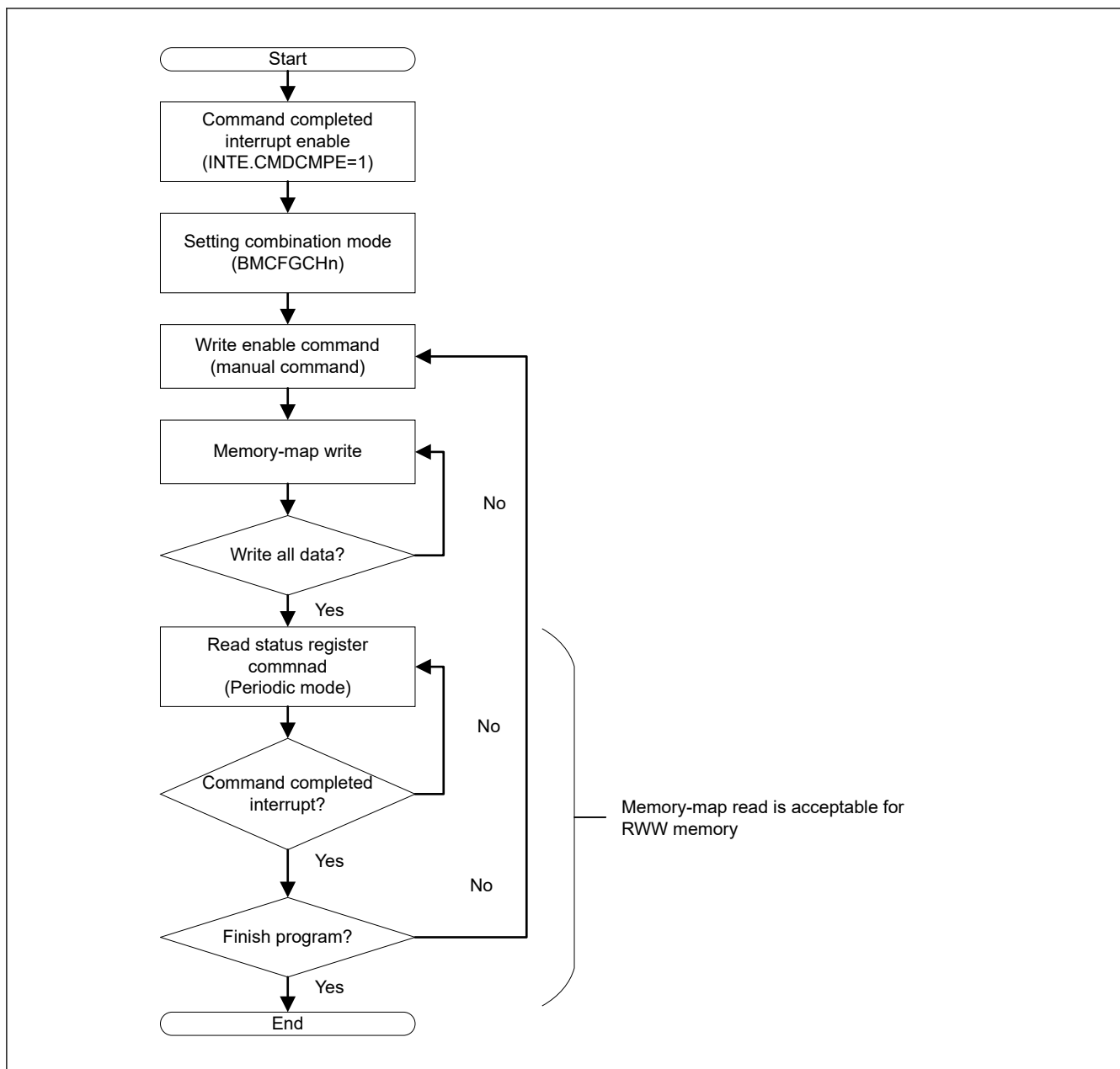


Figure 44.31 Flow of read while write

44.4 SiP Product Configuration

Flash memory ISSI IS25WX064-JWLE is packaged for SiP product.

- VCC2 = 1.70 V to 2.00 V
- OM_1_SCLK frequency:
 - SDR without OM_DQS: Up to 50 MHz
 - SDR with OM_DQS/DDR: Up to 133 MHz.

For details, see [section 60, Electrical Characteristics](#) and the datasheet of IS25WX064.

The following settings should be set before accessing the SiP Flash memory.

- Set OSPI (11100b) to the PmnPFS.PSEL[4:0] bits of P603 to P607 and PC00 to PC07
- Set the LVOCR.LVO1E bit to 1
- Set the PSARB.PSARB17 bit to 0 unless whole SiP Flash area is marked as Non-secure

- Set the end address of CS1
 - Set the RACTL.RAEN bit to 1
 - Set the MER register to 0x0871_0FF0 for 8 MB SiP-Flash product
 - Set the MER register to 0x0831_0FF0 for 4 MB SiP-Flash product
 - Set the RACTL.RAEN bit to 0
- Set the LIOCFGCS1.DDRSMPEX[3:0] bits
 The t_{CKHDSH} value of SiP Flash memory is $t_{PERIOD}/2 + 6[ns]$. When OM1_SCLK is 133 MHz, the DDRSMPEX[3:0] bits should be 0x2.

Figure 44.32 shows connection between the MCU and the SiP Flash memory. The external pull-up should be attached to the PUP pin.

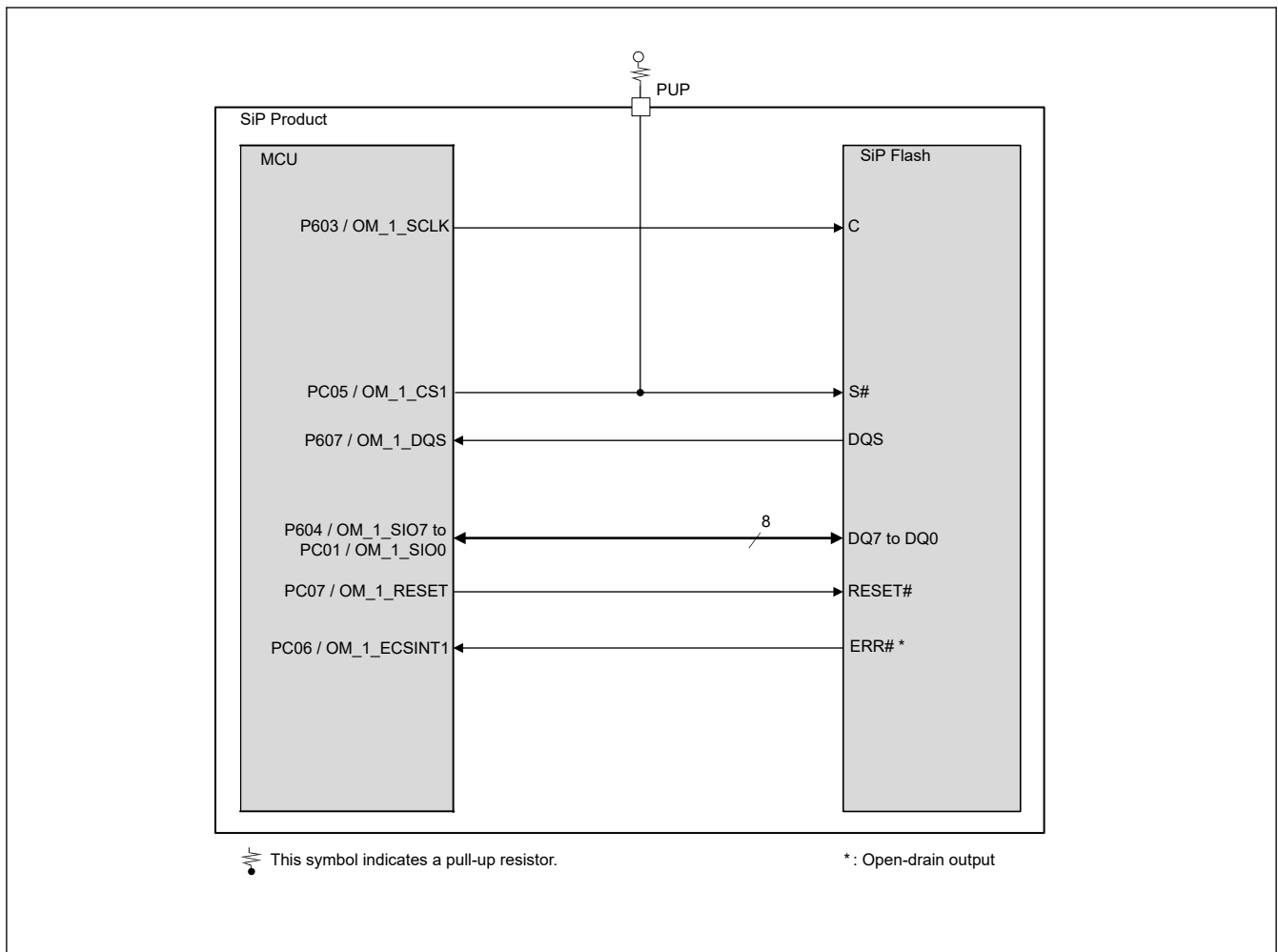


Figure 44.32 Connection between the MCU and the SiP Flash memory

44.5 Usage Notes

44.5.1 Software Standby Mode

Before transition to Software Standby mode, stop communication by Flow of Communication Stop, see [section 44.3.7.2. Flow of Communication Stop](#).

44.5.2 Constraints on Memory Access

Memory access has the following constraints:

- Not using the prefetch function if the application has access to the last 128-byte area

- Enable the prefetch function when read HyperFlash by DMAC/DTC in Block transfer mode or Repeat block transfer mode and source address is Incremented mode.

44.5.3 Constraint on Burst Length

AXI bus master must have a burst length of 16 or less.

44.5.4 Memory Write Combination Mode

When `BMCFGCHn.MWRCOMB = 1`, the memory write combination mode is enabled. However, the combination does not work if the bus master is CPU0 and OSPI data is under 32 bits. [Table 44.11](#) shows the possibility of a data write for each bus master.

Table 44.11 Data write possibility for each bus master

Bus master	Combination enable	Combination disable
CPU0 under 32-bit access	Not possible	Possible
CPU0 64-bit access	Possible	Possible
Other bus masters	Possible	Possible

44.5.5 Module-stop Function

OSPI operation can be disabled or enabled using Module Stop Control Register B (MSTPCRB). The OSPI module is initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see [section 11, Low Power Mode](#).

44.5.6 Byte Data Order in 8-pin DDR Mode

In 8-pin-DDR mode, OSPI read and write data in the order of byte 0 at the rising edge and byte 1 at the falling edge.

Some memory devices read and write in the swapped order in 8-pin DDR mode. For these devices, if 8-pin DDR read the data which is written in other than 8-pin DDR mode, this data should be written in swapped order.

44.5.7 Constraint on Prefetch Function

When all the following conditions are satisfied, prefetch function has no effect on the read access even if the PREEN bit is set to 1.

- Master is CPU0 or CPU1
- Memory attribution of OSPI region is set to cacheable memory by MPU in the CPU
- CPU cache is enabled
- DOTF is enabled.

44.5.8 Constraint on Block Protect setting for Flash 4MB SiP Product

Flash memory ISSI IS25WX064-JWLE has block protect function. There are some prohibited settings for Flash 4MB SiP Product (see [section 1.3. Part Numbering](#)).

[Table 44.12](#) shows block protect setting of TB and BP[3:0] bits in Status Register.

Table 44.12 Block protect setting for Flash 4MB SiP Product (1 of 2)

Status Register Bits					Byte Protected
TB	BP3	BP2	BP1	BP0	
1	0	0	0	0	0 KB
1	0	0	0	1	128 KB
1	0	0	1	0	256 KB
1	0	0	1	1	512 KB
1	0	1	0	0	1 MB

Table 44.12 Block protect setting for Flash 4MB SiP Product (2 of 2)

Status Register Bits					Byte Protected
TB	BP3	BP2	BP1	BP0	
1	0	1	0	1	2 MB
1	0	1	1	0	4 MB
Others: Setting prohibited					

44.5.9 Constraint on OTP Area for SiP Product

Flash memory ISSI IS25WX064-JWLE has 64-byte OTP area, but it is not supported in the SiP product.